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PURE MATHEMATICS WITH STATISTICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH STATISTICS

CHAPTER 6

NUMERICAL METHODS

SUGGESTED TIME: 5 PERIODS

IN THIS CHAPTER

This chapter introduces numerical methods for solving equations that cannot be solved exactly: locating roots by sign changes, then refining them with the bisection method and the Newton–Raphson iteration. Students learn to control accuracy, justify convergence, and apply these techniques to GCE-style modelling problems.

- By the end of this chapter I can explain why numerical methods are needed and what an approximate root is.
- By the end of this chapter I can locate a root of f(x) = 0 in an interval using a change of sign of a continuous function.
- By the end of this chapter I can use tables of values and graphs to bracket roots and count the roots of an equation.
- By the end of this chapter I can rearrange an equation into the form x = g(x) and use simple iteration x_n+1 = g(x_n) where appropriate.
- By the end of this chapter I can carry out the bisection method step by step and state the interval containing the root after n steps.
- By the end of this chapter I can apply the Newton–Raphson formula x_n+1 = x_n − f(x_n)/f'(x_n) to find a root to a given accuracy.

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Introduction and Location of Roots
2. The Bisection Method
3. The Newton–Raphson Method
4. Accuracy, Comparison and Applications
5. Integration activity
6. Methods and worked examples
7. Exercises
8. Solutions to the exercises
9. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can explain why numerical methods are needed and what an approximate root is.
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- By the end of this chapter I can apply the Newton–Raphson formula $x_{n+1} = x_n - f(x_n)/f'(x_n)$ to find a root to a given accuracy.
- By the end of this chapter I can interpret iteration geometrically using tangent lines and staircase/cobweb diagrams.
- By the end of this chapter I can decide when a numerical method fails or converges slowly and choose a suitable starting value.
- By the end of this chapter I can determine the accuracy of an approximation (decimal places, error bounds) and apply numerical methods to real-life problems.

PREREQUISITES

- Differentiation of polynomials, exponentials, logarithms and trigonometric functions (Lower Sixth calculus)
- Continuity and the behaviour of graphs of functions, including sketching curves
- Solving equations algebraically and manipulating functions (factorisation, rearrangement)
- Use of a scientific calculator in radians and degrees, and correct rounding to given decimal places
- Sequences and the idea of a limit (recurrence relations)

1 INTRODUCTION AND LOCATION OF ROOTS

A water-supply engineer in Bafoussam must size a pipe so that the flow equation $x^3 - 7x + 3 = 0$ is satisfied, yet no algebraic formula gives the answer. A mobile-money agent in Douala wants to know the exact interest rate hidden in a repayment schedule — again, the equation cannot be solved exactly. In real life, most equations that matter have no neat closed-form solution. Numerical methods give us powerful, systematic ways to find roots to any desired accuracy, using only arithmetic. This chapter shows how a simple calculator and a clever algorithm can solve problems that defeat pure algebra.

1.1 Why Numerical Methods?

From Lower Sixth onwards you have solved equations such as $x^2 - 5x + 6 = 0$ (factorisation or the quadratic formula) and $\sin x = \frac{1}{2}$ (standard values). But try solving any of these:

- $x^3 - x - 2 = 0$ (a cubic with no rational root);
- $e^x = 3x$ (mixing an exponential with a linear term);
- $\cos x = x$ (mixing a trigonometric function with a polynomial).

No algebraic manipulation will ever produce an exact answer in terms of surds, fractions or standard constants. Such equations are called **transcendental** or simply non-algebraic equations. Yet each of them does have solutions, and in engineering, finance, physics and statistics we genuinely need those solutions — often to 4, 6 or 10 decimal places.

DEFINITION **Root of an equation**

A **root** of the equation $f(x) = 0$ (also called a **zero** of the function f) is a number α such that $f(\alpha) = 0$. Geometrically, the roots of $f(x) = 0$ are the x-coordinates of the points where the curve $y = f(x)$ meets the x-axis.

DEFINITION **Approximate root and tolerance**

An **approximate root** of $f(x) = 0$ is a number x_0 which is close to the true root α. We say x_0 approximates α with tolerance ε (a small positive number) if

$$|x_0 - \alpha| < \varepsilon.$$

For example, stating a root "to 2 decimal places" means the error is at most 0.005; the tolerance is $\varepsilon = 0.005$.

WARNING **Exact versus approximate**

Writing $x = 1.521$ as "the solution" of $x^3 - x - 2 = 0$ is sloppy: it is an approximation. The correct statement is "the root is 1.521 correct to 3 decimal places", meaning the true root α satisfies $1.5205 \le \alpha < 1.5215$. In the GCE examination, always state the accuracy of your answer.

The strategy of this whole chapter has two phases:

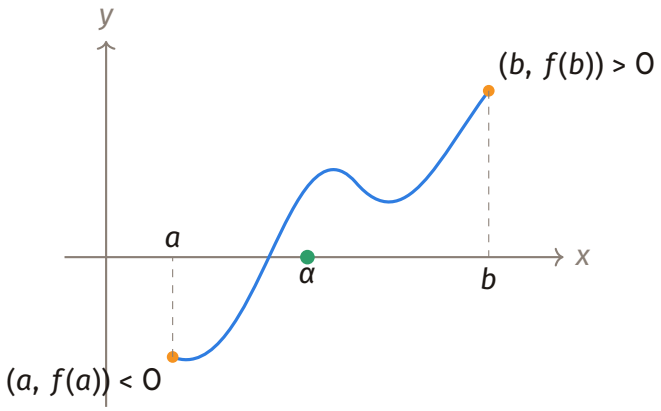
1. **Locate** the root: find an interval $[a, b]$ that is known to contain it (this section).
2. **Refine** the root: use an algorithm — bisection (Section 2) or Newton–Raphson (Section 3) — to shrink the interval or improve the estimate until the desired tolerance is reached.

1.2 The Intermediate Value Theorem: the Sign-Change Criterion

Suppose a function f is **continuous** on an interval $[a, b]$ — informally, its graph can be drawn from $(a, f(a))$ to $(b, f(b))$ without lifting the pen. If $f(a)$ is negative and $f(b)$ is positive, the curve starts below the x-axis and ends above it. Somewhere in between, it must cross the axis. That crossing point is a root.

PROPERTY **Intermediate Value Theorem — sign-change criterion**

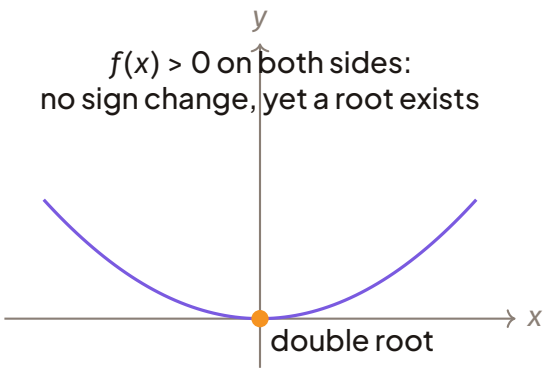
Let f be continuous on the closed interval $[a, b]$. If $f(a)$ and $f(b)$ have opposite signs (i.e. $f(a)\,f(b) < 0$), then the equation $f(x) = 0$ has at least one root in the open interval (a, b). (The full proof belongs to analysis and is not examinable; the statement and its use are.)



A continuous curve with $f(a) < 0$ and $f(b) > 0$ must cross the x-axis at least once between a and b.

WARNING **What the sign change does and does not say**

- A sign change proves at least one root in (a, b), but says nothing about uniqueness: there could be 3, 5, … roots (any odd number, counting multiplicity).
- No sign change does not prove there is no root: the interval could contain an even number of roots, or a double root where the curve is tangent to the axis.
- If f is not continuous on $[a, b]$, the test is unreliable: $f(x) = \frac{1}{x}$ gives $f(-1) = -1 < 0$ and $f(1) = 1 > 0$, yet $1/x = 0$ has no solution — the "sign change" comes from the discontinuity at $x = 0$, a false positive.



Counter-example: a curve tangent to the x-axis (double root) produces no sign change.

1.3 Locating Roots by a Table of Values

The most direct way to exploit the sign-change criterion is to evaluate $f(x)$ at integer points and look for a sign change between consecutive values.

METHOD **Locating roots with a table of values**

1. Write the equation in the form $f(x) = 0$.
2. Compute $f(x)$ at consecutive integers $x = \ldots, -2, -1, 0, 1, 2, \ldots$ over the required range.
3. Each sign change between n and $n + 1$ brackets a root in $(n, n + 1)$ — provided f is continuous there.
4. To locate the root more finely, tabulate f at steps of 0.1 inside that unit interval, then 0.01, and so on.
5. Check the graph (or further values) to make sure no root has been missed (even numbers of roots hide from the test).

WORKED EXAMPLE 1 **Locating the root of $x^3 - x - 2 = 0$**

Show that the equation $x^3 - x - 2 = 0$ has a root between 1 and 2, and locate it between two consecutive tenths.

SOLUTION

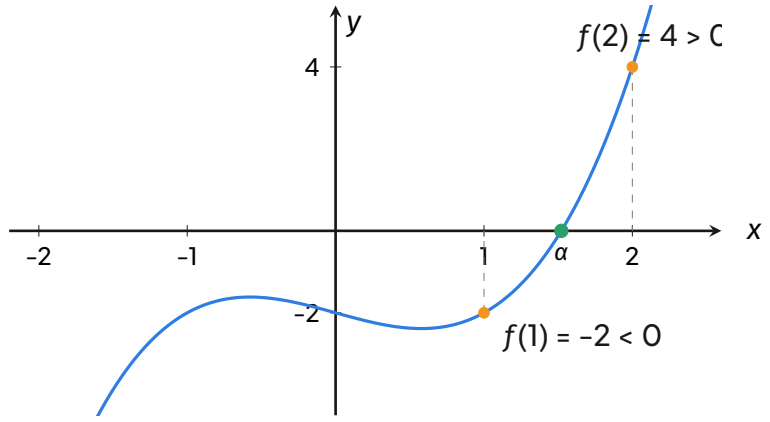
Let $f(x) = x^3 - x - 2$; f is a polynomial, hence continuous everywhere. Compute:

x	0	1	2	3
$f(x)$	-2	-2	4	22

Since $f(1) = -2 < 0$ and $f(2) = 4 > 0$, by the Intermediate Value Theorem there is a root $\alpha \in (1, 2)$. Now refine with steps of 0.1:

x	1.3	1.4	1.5	1.6	1.7
$f(x)$	-1.103	-0.656	-0.125	0.496	1.213

The sign change occurs between 1.5 and 1.6: hence $1.5 < \alpha < 1.6$. (In fact $\alpha \approx 1.5214$.)



Graph of $f(x) = x^3 - x - 2$: the sign change between $x = 1$ and $x = 2$ brackets the root $\alpha \approx 1.5214$.

1.4 Graphical Location of Roots

A quick sketch often reveals how many real roots an equation has and roughly where they are — essential information before starting any numerical algorithm, since the table method can miss roots.

Method 1: sketch $y = f(x)$ directly. The roots are the x-intercepts. This works well when f is easy to sketch.

Method 2: split into two familiar curves. Rearrange $f(x) = 0$ as $g(x) = h(x)$ where g and h are standard graphs (lines, parabolas, exponentials, trigonometric curves). The roots of $f(x) = 0$ are the x-coordinates of the intersection points of $y = g(x)$ and $y = h(x)$.

METHOD **Locating roots graphically by splitting**

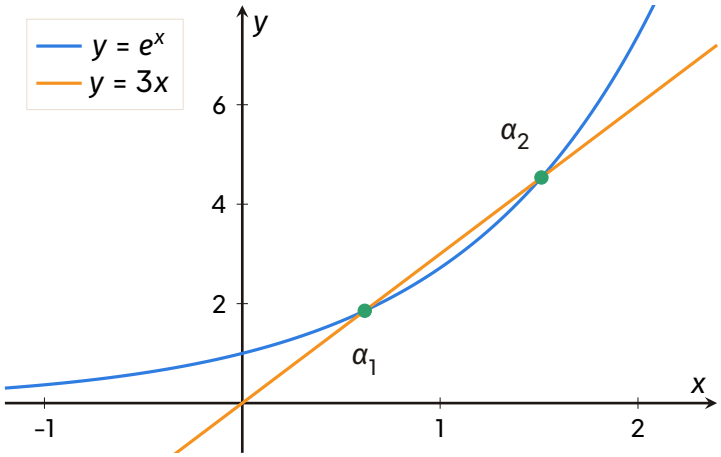
1. Rearrange the equation into the form $g(x) = h(x)$ with g, h easy to sketch.
2. Sketch $y = g(x)$ and $y = h(x)$ on the same axes (a careful freehand sketch is enough).
3. Count the intersection points: this gives the number of real roots.
4. Read off approximate positions and choose integer bounds $n, n + 1$ bracketing each root.
5. Confirm each bracket by a sign change of $f(x) = g(x) - h(x)$.

EXEMPLE **The equation $e^x = 3x$**

Take $g(x) = e^x$ and $h(x) = 3x$. The exponential curve and the straight line meet twice, so $e^x = 3x$ has exactly two real roots. Reading the graph, one root lies between 0 and 1, the other between 1 and 2. Check with $f(x) = e^x - 3x$:

$$f(0) = 1 > 0, \quad f(1) = e - 3 \approx -0.28 < 0, \quad f(2) = e^2 - 6 \approx 1.39 > 0.$$

Both sign changes are confirmed: roots in $(0, 1)$ and $(1, 2)$.



The curves $y = e^x$ and $y = 3x$ intersect twice: the equation $e^x = 3x$ has two real roots, $\alpha_1 \in (0, 1)$ and $\alpha_2 \in (1, 2)$.

WORKED EXAMPLE 2 **Counting and bracketing all roots of $\cos x = x$ and of $x^3 - 7x + 3 = 0$**

(a) Show that $\cos x = x$ has exactly one real root and bracket it between consecutive integers. (b) Show that $x^3 - 7x + 3 = 0$ has three real roots and give integer bounds for each.

SOLUTION

(a) Sketch $y = \cos x$ (oscillating between -1 and 1) and $y = x$ (a straight line through the origin). For $x > 1$ the line is above the cosine's maximum; for $x < -1$ it is below its minimum. They cross exactly once, in $(0, 1)$. Check with $f(x) = \cos x - x$ (x in radians):

$$f(0) = 1 > 0, \qquad f(1) = \cos 1 - 1 \approx 0.5403 - 1 = -0.4597 < 0.$$

Sign change confirmed: the unique root lies in $(0, 1)$.
(b) Let $f(x) = x^3 - 7x + 3$, continuous everywhere. Tabulate at integers:

x	-3	-2	-1	0	1	2	3	4
$f(x)$	-3	9	9	3	-3	-3	9	39

Sign changes occur between -3 and -2, between 0 and 1, and between 2 and 3. Since a cubic has at most three real roots, these are all of them:

$$\alpha_1 \in (-3, -2), \qquad \alpha_2 \in (0, 1), \qquad \alpha_3 \in (2, 3).$$

This is exactly the engineer's equation from Bafoussam — and we now know it has three real solutions, so a physical constraint will be needed to select the meaningful one.

WARNING **Traps in locating roots**

- Radians! When mixing trigonometric functions with polynomials ($\cos x = x$), your calculator must be in radian mode.
- A single table at unit steps can miss two roots lying in the same unit interval (e.g. roots at 1.2 and 1.8). Always combine the table with a sketch or with knowledge of the maximum number of roots (a polynomial of degree n has at most n real roots).
- Never apply the sign-change test across an asymptote: $f(x) = \dfrac{1}{x}$ "changes sign" between -1 and 1 but has no zero there.

1.5 First Ideas of Iteration

Locating a root between two integers is only the beginning. To refine it, numerical methods use **iteration**: starting from an estimate x_0, we apply a rule that produces a (hopefully better) estimate x_1, then x_2, and so on, generating a sequence (x_n).

One natural idea is to rearrange $f(x) = 0$ into the form

$$x = g(x)$$

and then compute

$$x_{n+1} = g(x_n), \qquad n = 0, 1, 2, \ldots$$

If the sequence (x_n) settles down to a limit ℓ, then (provided g is continuous) $\ell = g(\ell)$, so ℓ is a root of the rearranged equation. We say the iteration **converges**. If the terms drift away or oscillate wildly, it **diverges** — and the choice of rearrangement matters enormously.

EXEMPLE **A first taste of iteration:** $x^3 - x - 2 = 0$

Rearrange as $x = \sqrt[3]{x+2}$, i.e. $g(x) = (x+2)^{1/3}$. Starting from $x_0 = 1.5$:

$$x_1 = (1.5 + 2)^{1/3} \approx 1.5183,$$
$$x_2 = (1.5183 + 2)^{1/3} \approx 1.5211,$$
$$x_3 \approx 1.5214, \qquad x_4 \approx 1.5214.$$

The sequence converges rapidly to the root $\alpha \approx 1.5214$ found earlier. But the equally valid rearrangement $x = x^3 - 2$ gives $x_1 = 1.375$, $x_2 \approx 0.5996$, $x_3 \approx -1.7846$, $x_4 \approx -7.684$, … — divergence! Same equation, same starting point, different rearrangement, opposite behaviour.

KEY POINTS **Key points of this section**

- A **root** of $f(x) = 0$ is a value α with $f(\alpha) = 0$; most real-world equations can only be solved approximately, to a stated **tolerance**.
- Intermediate Value Theorem: if f is continuous on $[a, b]$ and $f(a)f(b) < 0$, then $f(x) = 0$ has at least one root in (a, b).
- A sign change proves existence, not uniqueness; absence of a sign change does not prove absence of roots (double roots, even numbers of roots); discontinuities can create false sign changes.
- Roots can be located by a table of values or graphically, either from $y = f(x)$ or by splitting into two intersecting curves $y = g(x)$, $y = h(x)$.
- Iteration $x_{n+1} = g(x_n)$ generates a sequence of estimates that may converge to a root — the engine behind the bisection and Newton–Raphson methods of the next sections.

EXERCISE 1 **Locating roots**

1. Show that the equation $x^3 + 2x - 5 = 0$ has exactly one real root and that it lies between 1 and 2. Locate it between two consecutive tenths.
2. By sketching $y = \ln x$ and $y = 2 - x$, show that $\ln x = 2 - x$ has exactly one root, and find

integer bounds for it.

3. Explain why the sign change of $f(x) = \dfrac{x+1}{x-2}$ between $x = 1$ and $x = 3$ does not guarantee a root of $f(x) = 0$ in $(1, 3)$. Where is the actual root?
4. Rearrange $x^2 - 5x + 3 = 0$ in the form $x = g(x)$ in two different ways and test, with $x_0 = 4$, which rearrangement seems to converge.

SOLUTION **Exercise 1**

1. $f(x) = x^3 + 2x - 5$ is continuous; $f(1) = -2 < 0$, $f(2) = 7 > 0$, so a root lies in $(1, 2)$. Since $f'(x) = 3x^2 + 2 > 0$ for all x, f is strictly increasing, so the root is unique. Refining: $f(1.3) \approx -0.203 < 0$, $f(1.4) \approx 0.544 > 0$, so the root lies in $(1.3, 1.4)$.
2. $y = \ln x$ is increasing from $-\infty$, $y = 2 - x$ is a decreasing line; they cross exactly once, for $x \in (1, 2)$ since at $x = 1$: $\ln 1 = 0 < 1$; at $x = 2$: $\ln 2 \approx 0.693 > 0$. With $f(x) = \ln x + x - 2$: $f(1) = -1 < 0$, $f(2) \approx 0.693 > 0$. Root in $(1, 2)$.
3. f is not continuous on $[1, 3]$: it has a vertical asymptote at $x = 2$. The sign change is a false positive caused by the discontinuity. The actual root satisfies $x + 1 = 0$, i.e. $x = -1$, which is outside $(1, 3)$.
4. (i) $x = \dfrac{x^2+3}{5}$: $x_0 = 4 \to x_1 = 3.8 \to x_2 \approx 3.488 \to \ldots$ converging towards 4.303? Checking: the roots are $\dfrac{5 \pm \sqrt{13}}{2} \approx 4.303$ and 0.697; the values decrease towards 4.303 — convergence. (ii) $x = 5 - \dfrac{3}{x}$: $x_0 = 4 \to x_1 = 4.25 \to x_2 \approx 4.294 \to x_3 \approx 4.301$ — also convergent to 4.303. (Other rearrangements such as $x = \sqrt{5x - 3}$ behave differently; the point is that convergence depends on the rearrangement.)

2 THE BISECTION METHOD

Having learned how to locate a root of a continuous function $f(x)$ by finding an interval $[a, b]$ where $f(a)f(b) < 0$, we now need a systematic procedure to approximate that root to any desired accuracy. The simplest and most robust such procedure is the **bisection method** (also called the interval-halving method). It repeatedly halves the interval, always keeping the subinterval where the sign change occurs, thereby trapping the root in an ever-shrinking interval.

2.1 Principle of the Bisection Method

Suppose f is continuous on $[a, b]$ and $f(a)f(b) < 0$. By the Intermediate Value Theorem, there is at least one root $\alpha \in (a, b)$. The idea is straightforward:

1. Compute the midpoint $m = \dfrac{a+b}{2}$.

2. Evaluate $f(m)$.
3. If $f(m) = 0$, we have found the exact root (rare in practice).
4. If $f(a)f(m) < 0$, the root lies in $[a, m]$; set $b = m$.
5. If $f(m)f(b) < 0$, the root lies in $[m, b]$; set $a = m$.

In either case, the new interval is exactly half the length of the previous one and still brackets the root. Repeating this process produces a sequence of nested intervals $[a_n, b_n]$ whose lengths tend to zero, forcing both endpoints to converge to the root α.

DEFINITION **Bisection Algorithm**

Given a continuous function f on $[a, b]$ with $f(a)f(b) < 0$ and a tolerance $\varepsilon > 0$:

1. Set $n = 0$, $a_0 = a$, $b_0 = b$.
2. While $b_n - a_n > \varepsilon$ (or until endpoints agree to required decimal places):
   - Compute $m_n = \dfrac{a_n + b_n}{2}$.
   - If $f(a_n)f(m_n) < 0$, set $a_{n+1} = a_n$, $b_{n+1} = m_n$.
   - Else, set $a_{n+1} = m_n$, $b_{n+1} = b_n$.
3. The approximate root is m_n (or $\frac{a_n+b_n}{2}$) with error $\leq \frac{b_n-a_n}{2}$.

METHOD **Step-by-Step Bisection Procedure**

1. Verify the bracket: Ensure f is continuous on $[a, b]$ and $f(a)f(b) < 0$.
2. Draw a table with columns: n, a_n, b_n, m_n, $f(m_n)$, Sign of $f(m_n)$, New Interval.
3. Iterate: Compute m_n, evaluate $f(m_n)$, check its sign against $f(a_n)$ (or $f(b_n)$), and write the new interval.
4. Stop: When the interval width $b_n - a_n$ is smaller than the required tolerance, or when a_n and b_n agree to the required number of decimal places.
5. State the root: The best estimate is the midpoint of the final interval. Give the answer to the requested accuracy.

2.2 Worked Example: Solving $x^3 - 7x + 3 = 0$ in $[0, 1]$

Let $f(x) = x^3 - 7x + 3$. We have $f(0) = 3 > 0$ and $f(1) = -3 < 0$, so a root lies in $[0, 1]$. We perform bisection until the root is determined correct to **two decimal places** (i.e., interval width < 0.005 or endpoints agree to 2 d.p.).

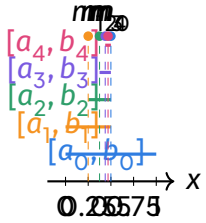
EXEMPLE **Bisection Iterations for $x^3 - 7x + 3 = 0$**

n	a_n	b_n	$m_n = \frac{a_n+b_n}{2}$	$f(m_n)$	Sign	New In
0	0	1	0.5	0.125 − 3.5 + 3 = −0.375	−	
1	0	0.5	0.25	0.015625 − 1.75 + 3 = 1.265625	+	[
2	0.25	0.5	0.375	0.052734 − 2.625 + 3 = 0.427734	+	[0
3	0.375	0.5	0.4375	0.083740 − 3.0625 + 3 = 0.021240	+	[0
4	0.4375	0.5	0.46875	0.103027 − 3.28125 + 3 = −0.178223	−	[0.43
5	0.4375	0.46875	0.453125	0.093018 − 3.171875 + 3 = −0.078857	−	[0.43
6	0.4375	0.453125	0.4453125	0.088287 − 3.117188 + 3 = −0.028901	−	[0.437
7	0.4375	0.4453125	0.44140625	0.085999 − 3.089844 + 3 = −0.003845	−	[0.437
8	0.4375	0.44140625	0.439453125	0.084869 − 3.076172 + 3 = 0.008697	+	[0.43945

After 8 iterations, the interval is $[0.439453125,\ 0.44140625]$. Its width is $0.001953125 < 0.005$. Both endpoints round to **0.44** to two decimal places. The midpoint of the final interval is $0.4404296875 \approx 0.44$.

KEY POINTS **Final Answer**

The root of $x^3 - 7x + 3 = 0$ in $[0, 1]$ correct to two decimal places is $\alpha = 0.44$.



Successive nested intervals in the bisection method. Each midpoint m_n splits the current interval; the half containing the root is kept.

2.3 Stopping Criteria and Error Bound

At each step the interval length is halved: $b_{n+1} - a_{n+1} = \frac{1}{2}(b_n - a_n)$. After n bisections, the length is

$$b_n - a_n = \frac{b_0 - a_0}{2^n}.$$

The true root α lies somewhere in $[a_n, b_n]$. If we take the midpoint m_n as our approximation, the

maximum possible error is half the interval length.

PROPERTY **Error Bound after n Bisections**

If the initial bracket is $[a, b]$ and n bisections have been performed, then for the midpoint $m_n = \frac{a_n+b_n}{2}$,

$$|\alpha - m_n| \le \frac{b_n - a_n}{2} = \frac{b-a}{2^{n+1}}.$$

This formula allows us to predict **how many iterations** are needed to guarantee a given accuracy before starting the computation.

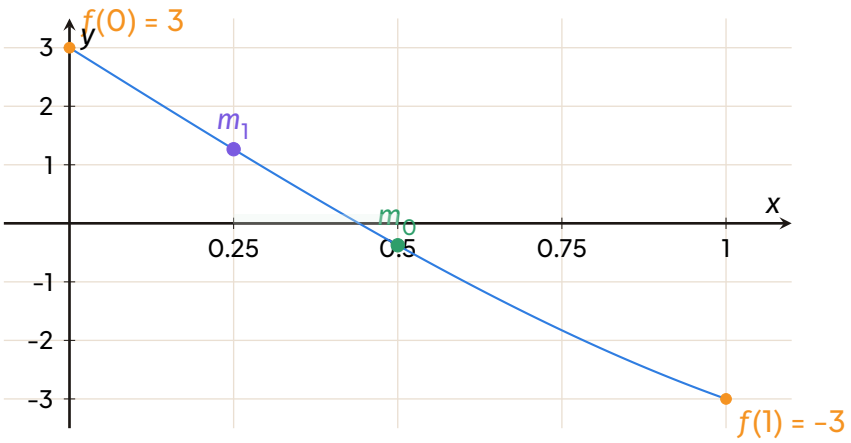
METHOD **Choosing the Number of Iterations for a Given Accuracy**

To achieve an error $< \varepsilon$ (or interval width $< \delta$):

1. For error bound: solve $\frac{b-a}{2^{n+1}} < \varepsilon$ for the smallest integer n.
2. For interval width: solve $\frac{b-a}{2^n} < \delta$ for the smallest integer n.
3. Equivalently: $n > \frac{\ln(b-a) - \ln(\delta)}{\ln 2}$ (width) or $n > \frac{\ln(b-a) - \ln(2\varepsilon)}{\ln 2}$ (error).

EXEMPLE **Predicting Iterations**

For $f(x) = x^3 - 7x + 3$ on $[0, 1]$, we want the root correct to 3 decimal places (error < 0.0005). Initial width $b - a = 1$. We need $\frac{1}{2^{n+1}} < 0.0005 \Rightarrow 2^{n+1} > 2000 \Rightarrow n + 1 > \log_2 2000 \approx 10.97$. So $n \ge 10$ iterations guarantee the required accuracy. (In practice, we stop as soon as the endpoints agree to 3 d.p., which may happen slightly earlier.)



Graph of $f(x) = x^3 - 7x + 3$ showing the first two midpoints $m_0 = 0.5$ and $m_1 = 0.25$. The shaded region $[0.25, 0.5]$ is the interval retained after two bisections.

2.4 Advantages and Disadvantages

Aspect	Advantages	Disadvantages
Convergence	**Guaranteed** for any continuous f with a sign change. No divergence possible.	**Slow** (linear convergence). Error halves each step; ~ 3.3 iterations per decimal digit.
Simplicity	Very easy to program/implement; only requires function evaluations (no derivative).	Requires a **valid starting bracket** $[a, b]$ with $f(a)f(b) < 0$.
Robustness	Unaffected by function shape (flat regions, multiple roots in interval — finds one).	Cannot find roots of even multiplicity (no sign change, e.g., $f(x) = x^2$ at 0).
Error Control	Exact error bound known at every step: $\frac{b_n - a_n}{2}$.	No way to accelerate; always one bit per iteration.

2.5 GCE Examination Technique

In the GCE Advanced Level examination, you will often be asked to:

- Perform a fixed number of bisections (e.g., "Carry out 4 iterations").
- Find the root to a stated accuracy (e.g., "correct to 3 decimal places").
- State the number of iterations needed to achieve a given tolerance.

WARNING **Common Mistakes to Avoid**

1. Sign errors in $f(m)$: One wrong sign sends you to the wrong half-interval, and all sub-

sequent rows are meaningless. Double-check every $f(m)$ calculation.

2. Using an endpoint as the final answer: The best estimate is the midpoint of the final interval, not a_n or b_n. The midpoint halves the maximum error.
3. Stopping too early: "Correct to 2 d.p." means the interval width must be < 0.005 or the endpoints must round to the same 2 d.p. value. Do not stop just because $f(m)$ is small.
4. Messy tables: Draw a neat table with a ruler. Show a_n, b_n, m_n, $f(m_n)$, and the new interval clearly. Examiners award marks for the table itself.
5. Forgetting to verify the initial bracket: Always state $f(a)$ and $f(b)$ and note $f(a)f(b) < 0$ before starting.

WARNING **The "Midpoint vs Endpoint" Trap**

Suppose after several iterations you have $[0.439, 0.441]$. The root is 0.44 to 2 d.p.

- Midpoint = 0.440 → rounds to 0.44. Correct.
- Left endpoint = 0.439 → rounds to 0.44. (Works here but not guaranteed.)
- Right endpoint = 0.441 → rounds to 0.44. (Works here but not guaranteed.)

If the interval were $[0.439, 0.440]$, midpoint = 0.4395 → 0.44, but right endpoint = 0.440 → 0.44, left = 0.439 → 0.44. Still okay. But if interval were $[0.434, 0.436]$ (width 0.002), midpoint = 0.435 → 0.44, endpoints give 0.43 and 0.44 — disagree. Always use the midpoint.

WORKED EXAMPLE 3 **GCE Style Question**

Statement: The function $f(x) = e^x - 3x$ has a root α in the interval $[0, 1]$.

1. Verify that α lies in $[0, 1]$.
2. Perform three iterations of the bisection method to approximate α.
3. Using your results, state the value of α correct to one decimal place, giving a reason for your answer.

SOLUTION

Detailed Solution:
1. Bracket verification: $f(0) = e^0 - 0 = 1 > 0$. $f(1) = e - 3 \approx 2.71828 - 3 = -0.28172 < 0$. f is continuous (exponential and polynomial), and $f(0)f(1) < 0$. Hence a root $\alpha \in (0, 1)$.

2. Three iterations: We need e^x values: $e^{0.5} \approx 1.64872$, $e^{0.75} \approx 2.11700$, $e^{0.625} \approx 1.86825$.

n	a_n	b_n	m_n	$f(m_n)$	Sign	New Interval
0	0	1	0.5	1.64872 - 1.5 = 0.14872	+	[0.5, 1]
1	0.5	1	0.75	2.11700 - 2.25 = -0.13300	-	[0.5, 0.75]
2	0.5	0.75	0.625	1.86825 - 1.875 = -0.00675	-	[0.5, 0.625]

3. Root to 1 d.p.: After 3 iterations, the interval is [0.5, 0.625]. Width = 0.125. The midpoint is $m_3 = 0.5625$. To 1 d.p., the endpoints are 0.5 and 0.6 (since 0.625 rounds to 0.6). They do not agree. The required interval width for 1 d.p. certainty is < 0.05. Here $0.125 > 0.05$. Conclusion: With only three iterations, the root cannot be stated reliably to one decimal place. More iterations are needed until the interval width is less than 0.05 (or endpoints agree to 1 d.p.).

DID YOU KNOW? **Historical Note**

The bisection method is essentially a binary search on a continuous domain. It was known to ancient mathematicians in geometric form (e.g., Archimedes' method of exhaustion for π). In the 19th century, Bernard Bolzano formalised the Intermediate Value Theorem, giving the method a rigorous foundation. Today, it is the "fail-safe" fallback in many numerical libraries (like Brent's method) because it never diverges.

EXERCISE 2 **Practice Problems**

1. Use the bisection method to find the root of $x^3 - 2x - 5 = 0$ in $[2, 3]$ correct to 2 decimal places. Show your table.
2. How many bisections are needed to guarantee an error $< 10^{-4}$ for a root initially bracketed in $[1, 4]$?
3. The function $f(x) = \sin x - x/2$ has a root in $[\pi/2, \pi]$. Perform 4 iterations of bisection. What is the best estimate and its maximum error?
4. Explain why the bisection method cannot find the root of $f(x) = x^2$ at $x = 0$, even though $f(0) = 0$.

SOLUTION **Exercise 1**

$f(x) = x^3 - 2x - 5$. $f(2) = 8 - 4 - 5 = -1 < 0$, $f(3) = 27 - 6 - 5 = 16 > 0$. Root in $[2, 3]$. Target: 2

d.p. ⇒ width < 0.005 or endpoints agree to 2 d.p. Iterations (values to 6 d.p.):

n	a_n	b_n	m_n	$f(m_n)$	Sign	New Interval
0	2	3	2.5	5.625	+	[2, 2.5]
1	2	2.5	2.25	1.890625	+	[2, 2.25]
2	2	2.25	2.125	0.341797	+	[2, 2.125]
3	2	2.125	2.0625	-0.345703	-	[2.0625, 2.125]
4	2.0625	2.125	2.09375	-0.009277	-	[2.09375, 2.125]
5	2.09375	2.125	2.109375	0.165253	+	[2.09375, 2.109375]
6	2.09375	2.109375	2.1015625	0.077408	+	[2.09375, 2.1015625]
7	2.09375	2.1015625	2.09765625	0.033926	+	[2.09375, 2.09765625]
8	2.09375	2.09765625	2.095703125	0.012278	+	[2.09375, 2.095703125]
9	2.09375	2.095703125	2.0947265625	0.001494	+	[2.09375, 2.0947265625]
10	2.09375	2.0947265625	2.09423828125	-0.003894	-	[2.09423828125, 2.0947265625]

After 10 iterations, interval width = 0.000488 < 0.005. Endpoints 2.0942 and 2.0947 both round to 2.09. Root ≈ 2.09 to 2 d.p.

SOLUTION **Exercise 2**

Initial width $b - a = 3$. Error bound after n iterations: $\frac{3}{2^{n+1}} < 10^{-4}$. $2^{n+1} > 30000$. $2^{14} = 16384$, $2^{15} = 32768$. So $n + 1 \geq 15 \Rightarrow n \geq 14$. 14 iterations guarantee error $< 10^{-4}$.

SOLUTION **Exercise 3**

$f(x) = \sin x - x/2$. $[\pi/2, \pi] \approx [1.5708, 3.1416]$. $f(\pi/2) = 1 - \pi/4 \approx 0.2146 > 0$. $f(\pi) = 0 - \pi/2 \approx -1.5708 < 0$.

n	a_n	b_n	m_n	$f(m_n)$	Sign	New Interval
0	$\pi/2$	π	$3\pi/4 \approx 2.3562$	-0.4710	-	$[\pi/2, 3\pi/4]$
1	$\pi/2$	$3\pi/4$	$5\pi/8 \approx 1.9635$	-0.0578	-	$[\pi/2, 5\pi/8]$
2	$\pi/2$	$5\pi/8$	$9\pi/16 \approx 1.7671$	0.0972	+	$[9\pi/16, 5\pi/8]$
3	$9\pi/16$	$5\pi/8$	$19\pi/32 \approx 1.8653$	0.0242	+	$[19\pi/32, 5\pi/8]$

Best estimate: midpoint of final interval = $\frac{19\pi/32 + 5\pi/8}{2} = \frac{39\pi}{64} \approx 1.9144$. Maximum error = $\frac{5\pi/8 - 19\pi/32}{2} = \frac{\pi/32}{2} = \frac{\pi}{64} \approx 0.0491$.

SOLUTION **Exercise 4**

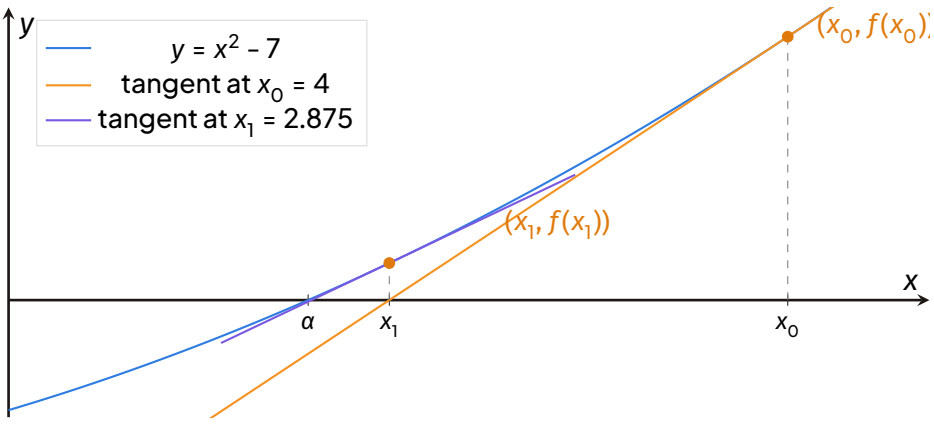
$f(x) = x^2$. $f(0) = 0$. For any interval $[a, b]$ containing 0 but not having 0 as an endpoint, $f(a) > 0$ and $f(b) > 0$ (since $x^2 > 0$ for $x \neq 0$). Thus $f(a)f(b) > 0$, no sign change. The Intermediate Value Theorem condition fails. The method relies on a sign change to bracket the root; a root of even multiplicity (touching the axis) does not produce a sign change, so bisection cannot "see" it.

3 THE NEWTON–RAPHSON METHOD

The bisection method of the previous section is safe and reliable: it never fails when a sign change has been located. But it is slow — each step gains only one binary digit, and reaching four decimal places typically requires more than a dozen iterations. We now study a much faster method, due to Isaac Newton and Joseph Raphson, which uses not only the values of f but also its slope. When it works, it is spectacularly efficient; when it fails, it fails badly — so we must also learn when and why it can go wrong.

3.1 The Geometric Idea

Suppose we want to solve $f(x) = 0$ and we have an estimate x_0 of the root α. Instead of working with the curve $y = f(x)$ itself, which may be complicated, we replace it near x_0 by the simplest curve we fully understand: the tangent line to the graph at the point $(x_0, f(x_0))$. The tangent is a good local approximation of the curve, so the point where the tangent crosses the x-axis should be close to the point where the curve crosses the x-axis — that is, close to the root. We call this intercept x_1, then repeat the process: draw the tangent at $(x_1, f(x_1))$, take its intercept x_2, and so on.



Newton–Raphson applied to $f(x) = x^2 - 7$. The tangent at $x_0 = 4$ meets the x-axis at $x_1 = 2.875$; the tangent there meets it at $x_2 \approx 2.6549$, already very close to $\alpha = \sqrt{7} \approx 2.6458$.

3.2 Derivation of the Formula

The tangent line to $y = f(x)$ at the point $(x_n, f(x_n))$ has gradient $f'(x_n)$, so its equation is

$$y - f(x_n) = f'(x_n)(x - x_n).$$

The next estimate x_{n+1} is where this tangent crosses the x-axis, i.e. where $y = 0$:

$$-f(x_n) = f'(x_n)(x_{n+1} - x_n) \quad \Longrightarrow \quad x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)}.$$

PROPERTY **Newton–Raphson iteration**

To solve $f(x) = 0$, choose a starting value x_0 and iterate

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}} \qquad (n = 0, 1, 2, \dots)$$

provided $f'(x_n) \neq 0$. The derivation from the tangent line is examinable and takes only two lines, as shown above.

METHOD **Applying the Newton–Raphson method**

1. Write the equation in the form $f(x) = 0$ and compute $f'(x)$ carefully.
2. Choose x_0 sensibly: the midpoint (or an endpoint) of a sign-change interval, or a value read from a graph of $y = f(x)$.
3. Compute $x_1, x_2, \dots$ with the formula, keeping at least two more decimal places than the accuracy required.
4. Stop when two successive iterates agree to the required accuracy, and verify the answer (e.g. by a sign change of f across the claimed rounded value).

WARNING **Differentiate first, differentiate right**

Every iterate depends on f'. If your derivative is wrong, every single value you compute afterwards is wrong, even if the arithmetic is perfect. Before iterating, check f' on a simple value of x.

WARNING **Radians!**

Whenever f involves trigonometric functions, your calculator must be in radian mode. The derivative of $\cos x$ is $-\sin x$ only when x is measured in radians.

3.3 Worked Examples

WORKED EXAMPLE 4 **Computing $\sqrt{7}$ to 4 decimal places**

Solve $x^2 - 7 = 0$ by Newton–Raphson and hence find $\sqrt{7}$ to 4 d.p.

SOLUTION

$f(x) = x^2 - 7$, so $f'(x) = 2x$ and the iteration simplifies to

$$x_{n+1} = x_n - \frac{x_n^2 - 7}{2x_n} = \frac{1}{2}\left(x_n + \frac{7}{x_n}\right).$$

Since $f(2) = -3 < 0$ and $f(3) = 2 > 0$, the root lies in $(2, 3)$; take $x_0 = 3$.

n	x_n	$f(x_n)$	x_{n+1}
0	3.000000	2.000000	2.666667
1	2.666667	0.111111	2.645833
2	2.645833	0.000434	2.645751
3	2.645751	-0.0000001	2.645751

x_3 and x_4 (both 2.645751) agree to 4 d.p., so $\sqrt{7} \approx 2.6458$. Note the speed: four iterations suffice, where bisection would need about thirteen.

WORKED EXAMPLE 5 **Solving $e^x - 3x = 0$**

Show that $e^x - 3x = 0$ has a root near 1.5 and find it to 4 d.p.

SOLUTION

$f(x) = e^x - 3x$; $f(1.5) = 4.4817 - 4.5 = -0.0183 < 0$ and $f(1.6) = 4.9530 - 4.8 = 0.1530 > 0$: sign change, so a root lies in $(1.5, 1.6)$. Take $x_0 = 1.5$. With $f'(x) = e^x - 3$:

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	1.500000	-0.018311	1.481689	1.512358
1	1.512358	0.000336	1.537410	1.512139
2	1.512139	-0.000001	1.537135	1.512139

The iterates agree to 4 d.p.: the root is 1.5121. (The equation has a second root near 0.6190; a different x_0 finds it.)

WORKED EXAMPLE 6 **Solving** $\cos x = x$

Find, to 4 d.p., the root of $\cos x - x = 0$ (calculator in radians).

SOLUTION

$f(x) = \cos x - x$, $f'(x) = -\sin x - 1$. Since $f(0) = 1 > 0$ and $f(1) = -0.4597 < 0$, take $x_0 = 0.75$.

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	0.750000	-0.018311	-1.681639	0.739111
1	0.739111	-0.000026	-1.673612	0.739095
2	0.739085	0.000000	-1.673577	0.739085

The root is 0.7391 (4 d.p.). This number is the unique fixed point of cos: pressing cos repeatedly on a radian-mode calculator converges to it!

3.4 Speed of Convergence

PROPERTY **Quadratic convergence**

Near a **simple root** α (i.e. $f(\alpha) = 0$, $f'(\alpha) \neq 0$), with a good starting value, Newton–Raphson converges quadratically: the error roughly squares at each step, so the number of correct decimal places approximately doubles at each iteration. (Proof not required at GCE.)

Look at the $\sqrt{7}$ table: the errors are roughly 3.5×10^{-1}, 2.1×10^{-2}, 8×10^{-5}, 10^{-9} — each error is proportional to the square of the previous one. This is why Newton–Raphson is the standard method built into calculators and computers.

DID YOU KNOW? **Historical note**

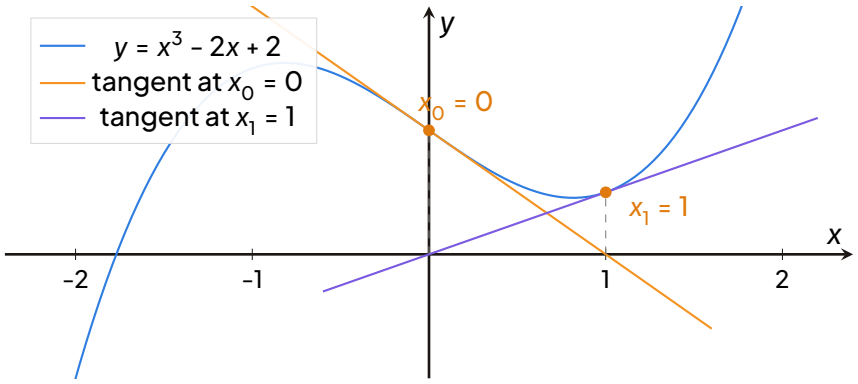
Isaac Newton described a version of this method in De analysi per aequationes numero terminorum infinitas (1669), but only for polynomials and without the general derivative formula. Joseph Raphson published the general iterative form $x_{n+1} = x_n - f(x_n)/f'(x_n)$ in Analysis aequationum universalis (1690). The method is therefore rightly named after both.

3.5 When Newton–Raphson Fails

WARNING **Failure cases — always check convergence**

- Horizontal tangent: if $f'(x_n) = 0$ (or is tiny), the formula divides by (nearly) zero and x_{n+1} is undefined or thrown far away. This happens near stationary points.
- Poor x_0: starting far from the root can send the iteration to a different root, into a cycle, or off to infinity.
- Oscillation: for $f(x) = x^3 - 2x + 2$ with $x_0 = 0$, one gets $x_1 = 1$, then $x_2 = 0$, $x_3 = 1, \ldots$ — an endless loop that never converges.
- Roots at inflection points (e.g. $f(x) = x^3$ at 0, where $f'(0) = 0$): convergence, when it occurs, is no longer quadratic.

Habit: watch the sequence of iterates. If they do not settle down, change x_0 or fall back on bisection.



Failure by oscillation: for $f(x) = x^3 - 2x + 2$, the tangent at $x_0 = 0$ gives $x_1 = 1$, and the tangent at $x_1 = 1$ sends us back to $x_2 = 0$. The iteration cycles forever and never reaches the real root (near −1.77).

3.6 Link with Fixed-Point Iteration (Enrichment)

Newton–Raphson is a special case of **fixed-point iteration** $x_{n+1} = g(x_n)$, which solves $x = g(x)$.

Indeed, the Newton formula is $x_{n+1} = g(x_n)$ with

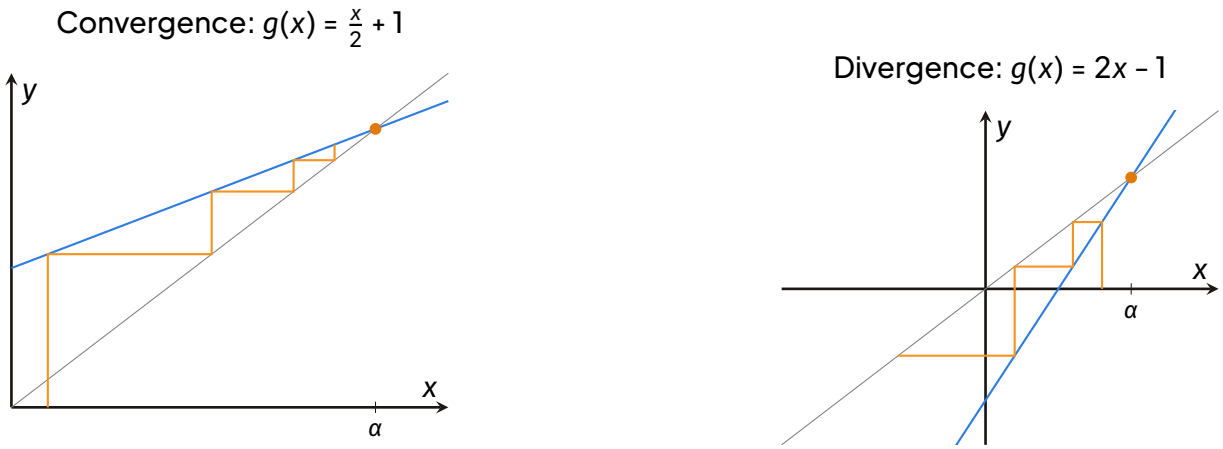
$$g(x) = x - \frac{f(x)}{f'(x)},$$

and one can check that $g'(\alpha) = 0$ at a simple root — the strongest possible case of the following rule.

PROPERTY **Convergence condition (enrichment)**

The iteration $x_{n+1} = g(x_n)$ converges to a fixed point α if x_0 is close enough to α and $|g'(x)| < 1$ near α. If $|g'(\alpha)| > 1$, it diverges. (Flagged as enrichment for strong candidates.)

Graphically, we plot $y = g(x)$ and $y = x$; the iteration appears as a staircase or cobweb: from x_n go vertically to the curve, then horizontally to the line $y = x$ — that gives x_{n+1}.



Cobweb diagrams. Left: $|g'| < 1$, the staircase spirals into the fixed point. Right: $|g'| = 2 > 1$, the cobweb moves away — the iteration diverges even though $x_0 = 0.8$ is close to $\alpha = 1$.

KEY POINTS **Key points**

- Newton–Raphson: $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$ — the next estimate is the x-intercept of the tangent at $(x_n, f(x_n))$.
- Choose x_0 from a sign-change interval or a graph; iterate until successive values agree; then verify.
- Convergence is quadratic near a simple root: correct digits roughly double each step.
- It fails when $f'(x_n) \approx 0$, when x_0 is badly chosen, or when the iteration cycles — always monitor the iterates.

- It is a fixed-point iteration with $g(x) = x - \frac{f(x)}{f'(x)}$; convergence of $x_{n+1} = g(x_n)$ requires $|g'| < 1$ near the root.

EXERCISE 3 **Practice**

1. Use Newton–Raphson with $x_0 = 2$ to find the root of $x^3 - 3x - 1 = 0$ in $(1, 2)$, to 4 d.p.
2. Show that $x^3 + x - 3 = 0$ has a root in $(1, 2)$ and find it to 4 d.p.
3. Explain, with a sketch, why Newton–Raphson with $x_0 = 1$ fails for $f(x) = x^3 - 3x + 5$.
4. (Enrichment) For $x_{n+1} = \cos x_n$, compute $g'(x)$ at the fixed point $\alpha \approx 0.7391$ and explain why the iteration converges.

SOLUTION **Exercise 1**

$f(x) = x^3 - 3x - 1$, $f'(x) = 3x^2 - 3$. Root in $(1, 2)$ since $f(1) = -3 < 0$, $f(2) = 1 > 0$. Start with $x_0 = 2$.

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	2.000000	1.000000	9.000000	1.888889
1	1.888889	0.072480	7.703704	1.879481
2	1.879481	0.000600	7.597322	1.879402
3	1.879402	≈ 0	7.596000	1.879402

x_3 and x_4 agree to 4 d.p. The root is 1.8794.

SOLUTION **Exercise 2**

$f(x) = x^3 + x - 3$, $f'(x) = 3x^2 + 1$. $f(1) = -1 < 0$, $f(2) = 7 > 0$: sign change in $(1, 2)$. Take $x_0 = 1.5$.

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	1.500000	1.875000	7.750000	1.258065
1	1.258065	0.249600	5.748100	1.214683
2	1.214683	0.007000	5.426200	1.213390
3	1.213390	0.000400	5.416900	1.213316
4	1.213316	≈ 0	5.416000	1.213316

x_4 and x_5 agree to 4 d.p. The root is 1.2133 (since 1.213316 rounds to 1.2133; checking

$f(1.2133) \approx 0.0001$, $f(1.2134) \approx 0.0005$, the root is 1.2133 to 4 d.p.).

SOLUTION **Exercise 3**

$f(x) = x^3 - 3x + 5$, $f'(x) = 3x^2 - 3$. At $x_0 = 1$, $f'(1) = 0$. The tangent at $(1, 3)$ is horizontal ($y = 3$) and never meets the x-axis, so x_1 is undefined. The method fails immediately.

SOLUTION **Exercise 4**

$g(x) = \cos x$, $g'(x) = -\sin x$. At the fixed point $\alpha \approx 0.7391$, $|g'(\alpha)| = \sin(0.7391) \approx 0.6736 < 1$. The convergence condition holds, so the cobweb spirals into the fixed point — convergence is guaranteed for x_0 sufficiently close to α.

4 ACCURACY, COMPARISON AND APPLICATIONS

In the previous section we met the Newton–Raphson method, a remarkably fast way of refining a root once we have located it. But speed alone is not enough for the GCE examiner: you must also be able to state precisely how accurate your answer is, to justify that accuracy, to choose the best method for a given problem, and to interpret the final number in a real situation. That is the business of this final section. By the end of it you will be able to take a full numerical-methods question from start to finish: locate, iterate, verify, interpret.

4.1 The Language of Accuracy

Decimal places versus significant figures.

Saying that $x = 2.0946$ is given **correct to 4 decimal places** means that the true value lies in the interval $[2.09455, 2.09465)$: we control the fifth decimal digit. Saying that $x = 0.003142$ is given **correct to 4 significant figures** means we control the fourth non-zero-leading digit. Decimal places measure absolute precision; significant figures measure relative precision. For roots of moderate size the two often coincide in practice, but examination questions about roots almost always ask for **decimal places**.

DEFINITION **Absolute error, tolerance, and "correct to k decimal places"**

Let α be an exact (unknown) value and x an approximation to it.

- The **absolute error** is $|x - \alpha|$.
- A **tolerance** (or error bound) is a number $\varepsilon > 0$ for which we can guarantee $|x - \alpha| \le \varepsilon$.

- We say x is **correct to k decimal places** if rounding α to k decimal places gives x; equivalently,
$$|x - \alpha| \le \tfrac{1}{2} \times 10^{-k}.$$
For example, "correct to 3 d.p." means an error of at most 0.0005.

WARNING **The calculator-display trap**

Writing down $x_4 = 1.5214$ because your calculator shows 1.5213797 ... does not prove that the root is 1.521 correct to 3 d.p. The iterates of a method may agree with each other while both are still wrong, or may agree only by luck. In the GCE, "show that the root is 1.521 correct to 3 decimal places" must be justified by a sign-change test (below) or by an error bound from the method itself. An unjustified rounded value earns no accuracy marks.

METHOD **Verifying a root correct to k decimal places by the sign-change test**

Suppose you believe the root α of a continuous function f is x_0 correct to k decimal places.

1. Form the **rounded interval** $[a, b] = [x_0 - \tfrac{1}{2} \times 10^{-k},\ x_0 + \tfrac{1}{2} \times 10^{-k}]$.
2. Compute $f(a)$ and $f(b)$ (keep plenty of guard digits).
3. If $f(a)$ and $f(b)$ have opposite signs, then by the Intermediate Value Theorem $\alpha \in (a, b)$, so α rounds to x_0 at k d.p. ■
4. If the signs are the same, your claimed value is not justified: iterate further and test again.

EXEMPLE **Justifying "correct to 3 d.p."**

We believe the root of $f(x) = x^3 - 2x - 5$ is 2.094 correct to 3 d.p. Test the interval [2.0935, 2.0945]:

$$f(2.0935) = 2.0935^3 - 2(2.0935) - 5 \approx -0.0020,$$
$$f(2.0945) = 2.0945^3 - 2(2.0945) - 5 \approx +0.0003.$$

There is a sign change on [2.0935, 2.0945], and f is continuous, so the root lies inside this interval. Hence $\alpha = 2.094$ correct to 3 d.p. This short computation is exactly what the examiner wants to see.

4.2 How Many Iterations? Predicting and Checking Convergence

Bisection: an exact count.

With the bisection method started on $[a, b]$, after n iterations the interval has length $(b - a)/2^n$, and the root is trapped inside it. Taking the midpoint as our estimate, the error is at most $\frac{b-a}{2^{n+1}}$. To guarantee accuracy ε we therefore need

$$\frac{b-a}{2^{n+1}} \le \varepsilon \quad \Longleftrightarrow \quad n \ge \frac{\ln\left(\frac{b-a}{2\varepsilon}\right)}{\ln 2}.$$

Each iteration buys roughly one extra binary digit; about 3.3 iterations are needed per decimal digit. This is why bisection is predictable but slow.

EXEMPLE **Counting bisection steps**

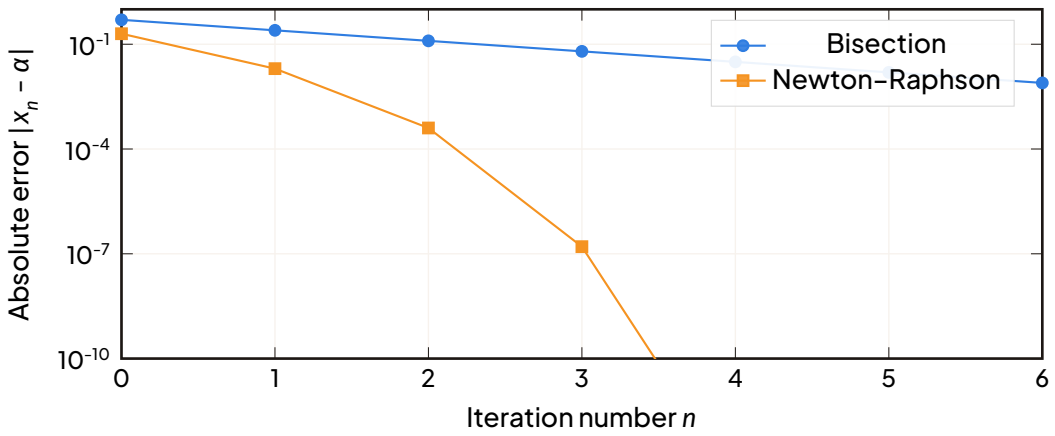
A root of $f(x) = x^3 + x - 3$ lies in $[1, 2]$. How many bisection iterations guarantee the midpoint is within 0.0005 of the root (i.e. 3 d.p. accuracy)?

$$n \ge \frac{\ln\left(\frac{1}{2 \times 0.0005}\right)}{\ln 2} = \frac{\ln 1000}{\ln 2} \approx \frac{6.9078}{0.6931} \approx 9.97.$$

So $n = 10$ iterations suffice (and 9 do not guarantee it).

Newton–Raphson: agreement of successive iterates.

Newton–Raphson typically doubles the number of correct digits at each step (quadratic convergence), but gives no built-in error bound. The practical test is: **stop when two successive iterates agree to the required number of decimal places**, then confirm with the sign-change test on the rounded interval. Agreement of x_n and x_{n+1} to k d.p. is strong evidence, but the sign change is the proof.



Typical convergence on the same equation (log scale): bisection halves the error each step, while Newton–Raphson roughly squares it — the error plunges after two or three steps.

4.3 Comparing the Methods and Choosing Wisely

PROPERTY **Comparison of the root-finding methods**

	Bisection	Newton–Raphson	Simple iteration $x = g(x)$
Speed	Slow (linear; error halved each step)	Very fast (quadratic near the root)	Linear; fast only if $\lvert g'(\alpha)\rvert$ is small
Reliability	Always converges if a sign change is bracketed	May diverge or cycle for a poor x_0; fails if $f'(x_n) = 0$	Converges only if $\lvert g'(x)\rvert < 1$ near the root
Requirements	Continuity and a sign change on $[a, b]$	Formula for $f'(x)$; good starting value	A rearrangement $x = g(x)$ with $\lvert g'\rvert < 1$
Error control	Exact bound $\frac{b-a}{2^{n+1}}$	Agreement of iterates + sign-change check	$\lvert x_{n+1} - x_n\rvert$ as a guide

(Stated for reference; the reasoning behind each row is examinable as commentary, not as a formal proof.)

METHOD **Choosing the right method in an examination**

1. Locate the root first with a sign change — this is compulsory whatever method fol-

lows.

2. If the question says "show that the root is … correct to …", use the sign-change test on the rounded interval.
3. If $f'(x)$ is easy to compute and a good x_0 is given, use Newton–Raphson: two or three iterations usually suffice.
4. If f' is awkward or unavailable, or the question imposes it, use bisection (or the given iterative formula).
5. If the question gives a rearrangement $x_{n+1} = g(x_n)$, use it as instructed and check $|g'| < 1$ if asked about convergence.

4.4 Application 1: Finance — the Interest Rate on a Loan

A trader in Douala borrows $P = 1\,000\,000$ FCFA and repays it by 12 monthly instalments of $M = 90\,000$ FCFA, the first one month after the loan. If i is the monthly interest rate, equating present values gives

$$P = M\,\frac{1-(1+i)^{-12}}{i}.$$

With $x = 1 + i$ this reduces to a polynomial equation; numerically it is cleaner to solve

$$f(i) = 90\,000\bigl(1-(1+i)^{-12}\bigr) - 1\,000\,000\,i = 0.$$

Since $f(0.01) \approx 90\,000(0.11255) - 10\,000 = 129.6 > 0$ and $f(0.02) \approx 90\,000(0.21151) - 20\,000 = -964.4 < 0$, the rate lies between 1% and 2% per month. Newton–Raphson with

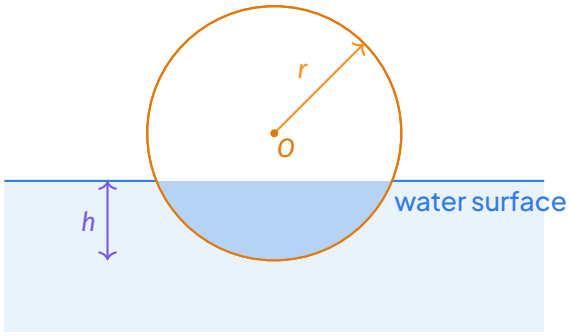
$$f'(i) = 1\,080\,000(1+i)^{-13} - 1\,000\,000, \qquad i_{n+1} = i_n - \frac{f(i_n)}{f'(i_n)},$$

starting from $i_0 = 0.012$:

$$\begin{aligned} i_1 &= 0.012 - \frac{-30.0}{-75\,520} \approx 0.011603,\\ i_2 &= 0.011603 - \frac{7.0}{-70\,120} \approx 0.011703,\\ i_3 &= 0.011703 - \frac{-3.0}{-69\,800} \approx 0.011660,\\ i_4 &\approx 0.011660. \end{aligned}$$

The iterates agree to 5 d.p.; a sign-change test on $[0.011655, 0.011665]$ confirms $i \approx 0.01166$ correct to 5 d.p., i.e. about 1.166% per month (approximately 14.9% per year). No algebraic formula exists for i — numerical methods are the only practical tool, which is exactly how banks and mobile-money platforms compute such rates.

4.5 Application 2: Engineering — Depth of a Floating Sphere



submerged depth h (unknown root)

A sphere of radius r floating in water: the submerged depth h is fixed by Archimedes' principle and is found by solving $f(h) = 0$.

A wooden sphere of radius r and relative density ρ (with $0 < \rho < 1$) floats in water. The volume of the submerged spherical cap of depth h is $V = \dfrac{\pi h^2(3r - h)}{3}$, and Archimedes' principle (weight of displaced water = weight of sphere) gives

$$\frac{\pi h^2(3r - h)}{3} = \rho \cdot \frac{4\pi r^3}{3} \quad \Longleftrightarrow \quad f(h) = h^3 - 3rh^2 + 4\rho r^3 = 0.$$

This cubic has no convenient exact solution, so we solve it numerically. Take $r = 10$ cm and wood of relative density $\rho = 0.6$: $f(h) = h^3 - 30h^2 + 2400$. Since $f(11) = 1331 - 3630 + 2400 = 101 > 0$ and $f(12) = 1728 - 4320 + 2400 = -192 < 0$, the root lies in $(11, 12)$. Newton–Raphson with $f'(h) = 3h^2 - 60h$ from $h_0 = 11$:

$$h_1 = 11 - \frac{101}{-297} \approx 11.34007,$$
$$h_2 = 11.34007 - \frac{0.59}{-294.61} \approx 11.34207,$$
$$h_3 = 11.34207 - \frac{-0.11}{-294.60} \approx 11.34170,$$
$$h_4 \approx 11.3418.$$

The iterates agree to 4 d.p.; checking $f(11.34125) > 0$ and $f(11.34175) < 0$ (sign change on the rounded interval for 3 d.p.) confirms $h = 11.342$ cm correct to 3 d.p. The same equation — with different constants — governs pipeline floats, buoys in the port of Kribi, and the calibration of fuel-tank gauges.

4.6 Application 3: Statistics Link — Calibrating a Probability Model

DID YOU KNOW? **Numerical methods inside statistics**

Many statistical questions secretly end with "solve $f(\theta) = 0$". For a Binomial model $X \sim B(n, p)$, finding the value of p that makes $P(X \geq 1)$ equal to a target, or finding the sample size n needed for a given confidence, leads to equations with no closed-form solution. Examiners love this crossover: it tests both your statistics and your numerical methods in one question.

WORKED EXAMPLE 7 **Finding a probability parameter numerically**

A machine fills sachets of water; the probability that a sachet is underweight is p. Quality control requires that in a batch of 20 sachets, $P(\text{at least one underweight}) = 0.5$. Find p correct to 4 decimal places.

SOLUTION

We need $1 - (1 - p)^{20} = 0.5$, i.e. $(1 - p)^{20} = 0.5$. (Here an exact solution $p = 1 - 0.5^{1/20}$ exists, but let us treat it as the examiner intends when the equation is not solvable exactly — the method is identical.) Set $f(p) = (1 - p)^{20} - 0.5$. Then

$$f(0.03) = 0.97^{20} - 0.5 \approx 0.5438 - 0.5 = 0.0438 > 0,$$

$$f(0.04) = 0.96^{20} - 0.5 \approx 0.4420 - 0.5 = -0.0580 < 0,$$

so $p \in (0.03, 0.04)$. Newton–Raphson with $f'(p) = -20(1 - p)^{19}$, starting at $p_0 = 0.035$:

$$p_1 = 0.035 - \frac{0.965^{20} - 0.5}{-20(0.965)^{19}} = 0.035 - \frac{-0.009877}{-10.158} \approx 0.034028,$$

$$p_2 = 0.034028 - \frac{0.965972^{20} - 0.5}{-20(0.965972)^{19}} \approx 0.034028 - \frac{0.0003}{-10.36} \approx 0.034057,$$

$$p_3 \approx 0.034064.$$

The iterates settle at $p \approx 0.03406$. Sign-change test on the rounded interval $[0.034055, 0.034065]$: $f(0.034055) > 0$ and $f(0.034065) < 0$, so $p = 0.0341$ correct to 4 d.p. Interpretation: the factory must keep the underweight rate at about 3.41% for the batch risk to be exactly one-half.

4.7 Full GCE-Style Synthesis

METHOD **Exam strategy: the four-move answer**

Every complete numerical-methods answer should show four moves, in this order:

1. Locate: exhibit a sign change of f on an interval $[a, b]$ (continuity + IVT).
2. Iterate: apply the required method, showing the formula, the substitution, and at least the first two iterates with full calculator values before rounding.
3. Verify: justify the final accuracy — sign change on the rounded interval $[x_0 \pm \frac{1}{2} \times 10^{-k}]$, or the bisection error bound.
4. Interpret: answer the question asked, in context, with units if any ("the depth is 11.342 cm", not just "$x = 11.342$").

WORKED EXAMPLE 8 **Complete synthesis question**

A spherical tank of radius 2 m lies horizontally. The volume of liquid of depth x metres in the tank is $V(x) = \dfrac{\pi x^2(6 - x)}{3}$. The tank contains 9 m^3 of liquid.

(a) Show that x satisfies $x^3 - 6x^2 + \dfrac{27}{\pi} = 0$, and that $1.3 < x < 1.4$.

(b) Use the Newton–Raphson method once, starting from $x_0 = 1.3$, to find a second approximation.

(c) Refine further and give x correct to 3 decimal places, justifying your answer.

SOLUTION

(a) Setting $V(x) = 9$: $\dfrac{\pi x^2(6 - x)}{3} = 9 \Rightarrow x^2(6 - x) = \dfrac{27}{\pi} \Rightarrow f(x) = x^3 - 6x^2 + \dfrac{27}{\pi} = 0$, where $\frac{27}{\pi} \approx 8.5944$.

$$f(1.3) = 2.197 - 10.14 + 8.5944 = 0.6514 > 0,$$
$$f(1.4) = 2.744 - 11.76 + 8.5944 = -0.4216 < 0,$$

so the root lies in $(1.3, 1.4)$.

(b) $f'(x) = 3x^2 - 12x$, so $f'(1.3) = 5.07 - 15.6 = -10.53$ and

$$x_1 = 1.3 - \frac{0.6514}{-10.53} \approx 1.36186.$$

(c) Continuing:

$$f(1.36186) \approx 2.5265 - 11.1275 + 8.5944 = -0.0066,$$
$$f'(1.36186) \approx 5.564 - 16.342 = -10.778,$$
$$x_2 = 1.36186 - \frac{-0.0066}{-10.778} \approx 1.36125,$$
$$x_3 \approx 1.36125.$$

The iterates agree to 5 d.p. Verification: on the rounded interval $[1.36075, 1.36125]$,

$$f(1.36075) \approx +0.0019 > 0, \qquad f(1.36125) \approx -0.0015 < 0,$$

a sign change, so $x = 1.361$ correct to 3 d.p. Interpretation: the liquid stands 1.361 m deep in the tank — the number a technician would mark on the dip-stick.

WARNING **Final warnings for the examination**

- Never claim "correct to 3 d.p." from the calculator display alone: justify with a sign change on $[x_0 - 0.0005,\ x_0 + 0.0005]$.
- Keep at least two guard digits during iteration; round only at the final answer.
- In Newton–Raphson, write the formula $x_{n+1} = x_n - f(x_n)/f'(x_n)$ explicitly — method marks depend on it.
- Check that your final answer lies in the physically meaningful range (a depth cannot exceed $2r$; a probability lies in $[0, 1]$; an interest rate is positive).
- If bisection is demanded, do not switch to Newton–Raphson halfway: follow the instructed method.

EXERCISE 4 **Practice**

1. A root of $f(x) = x^3 + x - 3$ is bracketed in $[1, 2]$. How many bisection iterations guarantee an error below 0.00005?
2. Show that the equation $e^{-x} = x$ has a root in $[0.5, 0.6]$. Use Newton–Raphson from $x_0 = 0.55$ to find the root correct to 4 decimal places, and justify the accuracy with a sign-change test.
3. A loan of 500 000 FCFA is repaid by 10 monthly payments of 53 000 FCFA. Write down the equation for the monthly rate i, show it has a root between 0.01 and 0.02, and find i correct to 4 decimal places.
4. For the floating sphere with $r = 5$ cm and $\rho = 0.5$, show that h satisfies $h^3 - 15h^2 + 250 = 0$ and find h correct to 3 decimal places. Why must the root with $h > 10$ be rejected?

SOLUTION **Exercise 1**

We need $\dfrac{b-a}{2^{n+1}} \le 0.00005$ with $b - a = 1$: $2^{n+1} \ge 20\,000$. Since $2^{14} = 16\,384 < 20\,000$ and $2^{15} = 32\,768 \ge 20\,000$, we need $n + 1 = 15$, i.e. $n = 14$ iterations.

SOLUTION **Exercise 2**

$f(x) = e^{-x} - x$: $f(0.5) = 0.6065 - 0.5 = 0.1065 > 0$; $f(0.6) = 0.5488 - 0.6 = -0.0512 < 0$: sign change, root in $(0.5, 0.6)$. With $f'(x) = -e^{-x} - 1$ and $x_0 = 0.55$: $x_1 = 0.55 - \dfrac{0.02695}{-1.57695} \approx 0.56709$; $x_2 \approx 0.56714$; $x_3 \approx 0.56714$. Test on $[0.567135, 0.567145]$: $f(0.567135) > 0$, $f(0.567145) < 0$, so $x = 0.5671$ correct to 4 d.p.

SOLUTION **Exercise 3**

Equation: $53\,000\left(1-(1+i)^{-10}\right) = 500\,000\,i$, i.e. $f(i) = 53\,000\left(1-(1+i)^{-10}\right) - 500\,000\,i$. $f(0.01) \approx 53\,000(0.09471) - 5\,000 = 19.8 > 0$; $f(0.02) \approx 53\,000(0.17965) - 10\,000 = -478.4 < 0$: root in $(0.01, 0.02)$. Newton–Raphson with $f'(i) = 530\,000(1+i)^{-11} - 500\,000$, starting at $i_0 = 0.011$:

$$i_1 = 0.011 - \frac{12.0}{-29\,890} \approx 0.011401,$$
$$i_2 = 0.011401 - \frac{-82.5}{-32\,010} \approx 0.011143,$$
$$i_3 = 0.011143 - \frac{-2.1}{-30\,800} \approx 0.011136,$$
$$i_4 \approx 0.011136.$$

Sign change on $[0.0111315, 0.0111365]$ confirms $i = 0.0111$ correct to 4 d.p., i.e. about 1.11% per month.

SOLUTION **Exercise 4**

Archimedes: $\frac{\pi h^2(15-h)}{3} = 0.5 \cdot \frac{4\pi(125)}{3} \Rightarrow h^2(15-h) = 250 \Rightarrow f(h) = h^3 - 15h^2 + 250 = 0$. $f(4) = 64 - 240 + 250 = 74 > 0$; $f(5) = 125 - 375 + 250 = 0$ exactly — so $h = 5.000$ cm (the sphere floats exactly half-submerged, as expected for $\rho = 0.5$). The cubic also has a root near $h \approx 13.7$, but $h > 2r = 10$ is physically impossible (the sphere would be more than fully submerged), so it is rejected: always match the mathematical roots to the physical context.

KEY POINTS **Key points of the section**

- "Correct to k d.p." means error $\le \frac{1}{2} \times 10^{-k}$; prove it by a sign change on $[x_0 - \frac{1}{2}10^{-k},\ x_0 + \frac{1}{2}10^{-k}]$.
- Bisection: guaranteed convergence, error bound $\frac{b-a}{2^{n+1}}$, slow. Newton–Raphson:

quadratic speed, needs f' and a good start, no built-in bound — verify afterwards.

- Exam structure: **locate** (sign change) → **iterate** (show formula and values) → **verify** (sign-change test) → **interpret** (context, units).
- Real problems — loan rates, floating bodies, probability calibration — reduce to $f(x) = 0$ and are solved exactly as in the exercises; reject non-physical roots.

5 INTEGRATION ACTIVITY

6 INTEGRATION ACTIVITY: THE MOBILE-MONEY INTEREST RATE MYSTERY

6.1 Aims of the Activity

This integration activity brings together all the skills developed in the chapter in a single, authentic financial problem. By the end of the activity, you should be able to:

- translate a real-life situation (a loan with fixed monthly repayments) into an equation of the form $f(r) = 0$ that cannot be solved algebraically;
- locate a root of $f(r) = 0$ by detecting a **change of sign** of f over an interval;
- apply the **bisection method** step by step, keeping track of the interval containing the root and of the maximum possible error;
- apply the **Newton–Raphson method** with the iteration $r_{n+1} = r_n - \dfrac{f(r_n)}{f'(r_n)}$ and control the number of decimal places guaranteed;
- verify the accuracy of a numerical answer by a final sign-change check;
- compare the speed of convergence of the two methods and communicate a conclusion in clear, non-technical language, as a consultant would.

Work in groups of three or four. Each group needs a scientific calculator. Keep at least six decimal places in all intermediate calculations, and only round at the very end.

6.2 Situation

The Mobile-Money Interest Rate Mystery.

Madame Eyenga runs a provisions store near the Mokolo market in Yaoundé. To restock her shop before the end-of-year festivities, she borrows 500 000 FCFA from a mobile-money lending service. The contract states that she will repay 115 000 FCFA at the end of each month, for exactly 5 months, after which the debt will be completely cleared.

The contract does not state the monthly interest rate r explicitly. However, the lender explains that each repayment of 115 000 FCFA first pays the interest due on the outstanding balance, and the remainder reduces the capital. A financial analyst can show (by discounting each of the five payments back to the date of the loan) that the monthly rate r satisfies the equation

$$500\,000\,r = 115\,000\left(1 - (1 + r)^{-5}\right).$$

Dividing both sides by 5 000, this becomes

$$100r = 23\left(1 - (1 + r)^{-5}\right).$$

This equation involves r both linearly and inside a power with a negative exponent: there is **no algebraic method** to solve it exactly. Madame Eyenga wants to know the monthly rate she is really being charged, correct to 4 decimal places (that is, to the nearest 0.01%), so that she can compare this offer with a bank loan.

Your group is hired as her advisory team. You must find the rate, justify its accuracy, and write her a short advisory note.

6.3 Tasks

1. **Setting up.** Define $f(r) = 100r - 23\left(1 - (1+r)^{-5}\right)$. Explain why solving the problem is equivalent to finding the root of $f(r) = 0$ in the interval $(0, 1)$, and state why an exact algebraic solution is impossible.

2. **Locating the root.** Compute $f(0.04)$ and $f(0.05)$, each to at least 4 decimal places. Explain, quoting the Intermediate Value Theorem, why there is a root r^* in the interval $(0.04, 0.05)$.

3. **Bisection.** Starting from the interval $[0.04, 0.05]$, carry out **three** steps of the bisection method. Present your work in a table with columns: step number, interval $[a_n, b_n]$, midpoint m_n, sign of $f(m_n)$, new interval. After three steps, state the interval containing r^*, its width, and the maximum error if you take the midpoint as your estimate.

4. **Newton–Raphson.** Show that $f'(r) = 100 - 115(1+r)^{-6}$. Starting from $r_0 = 0.05$, apply the Newton–Raphson iteration

$$r_{n+1} = r_n - \frac{f(r_n)}{f'(r_n)}$$

   until two successive iterates agree to 6 decimal places. Record $r_0, r_1, r_2, \ldots$ in a table with $f(r_n)$ and $f'(r_n)$.

5. **Accuracy check.** Let $\tilde{r}$ be your answer rounded to 4 decimal places. By evaluating f at $\tilde{r} - 0.00005$ and at $\tilde{r} + 0.00005$, confirm by a sign change that the root, rounded to 4 decimal places, is indeed $\tilde{r}$.

6. **Advisory note.** Write a short note (five to eight lines) to Madame Eyenga, in plain language, stating: the monthly interest rate as a percentage; how confident you are in this figure; and a comparison of the two methods (bisection versus Newton–Raphson) in terms of speed and reliability.

> **WARNING** **Traps to avoid**
>
> Do not confuse the monthly rate with an annual rate. In the bisection table, the new interval is chosen according to the sign of f at the midpoint, not its size. In Newton–Raphson, never round the iterates before the final answer: premature rounding can destroy the rapid convergence.

6.4 Worked Solutions

DEFINITION **Intermediate Value Theorem**

If a function f is continuous on a closed interval $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then there exists at least one $c \in (a, b)$ such that $f(c) = 0$.

WORKED EXAMPLE 9 **Task 1: Setting up the equation**

The loan equation $500\,000\,r = 115\,000\left(1 - (1 + r)^{-5}\right)$ simplifies, on division by $5\,000$, to

$$100r = 23\left(1 - (1 + r)^{-5}\right).$$

Define

$$f(r) = 100r - 23\left(1 - (1 + r)^{-5}\right).$$

The monthly rate r^* is precisely the value for which $f(r^*) = 0$. The unknown r appears both as a linear term $100r$ and inside the non-linear term $(1 + r)^{-5}$. No algebraic manipulation (factorisation, substitution, or use of standard functions) can isolate r; the equation is transcendental. Hence a numerical method is mandatory. As a monthly interest rate, r^* must lie between 0 and 1; since the total repayment is $5 \times 115\,000 = 575\,000$ FCFA (only 15% more than the principal), the rate is small, of the order of a few percent.

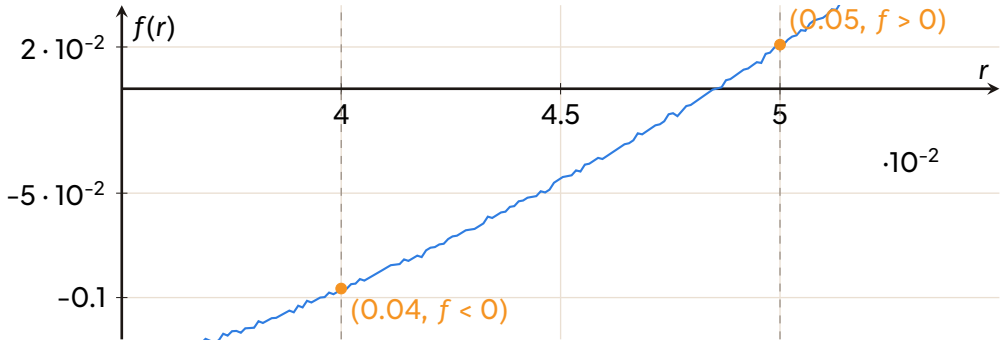
WORKED EXAMPLE 10 **Task 2: Locating the root by a sign change**

The function f is a sum of continuous functions (polynomial and power functions) on $(0, 1)$, therefore it is continuous on $[0.04, 0.05]$. We evaluate f at the endpoints, keeping six decimal places for safety:

$$\begin{aligned} f(0.04) &= 4 - 23\left(1 - 1.04^{-5}\right) \\ &= 4 - 23\,(1 - 0.821927) \\ &= 4 - 23 \times 0.178073 \\ &= 4 - 4.095679 = -0.095679 \approx -0.0957, \end{aligned}$$

$$\begin{aligned} f(0.05) &= 5 - 23\left(1 - 1.05^{-5}\right) \\ &= 5 - 23\,(1 - 0.783526) \\ &= 5 - 23 \times 0.216474 \\ &= 5 - 4.978902 = 0.021098 \approx 0.0211. \end{aligned}$$

Since f is continuous on $[0.04, 0.05]$ and $f(0.04) < 0 < f(0.05)$, the Intermediate Value Theorem guarantees the existence of a root $r^* \in (0.04, 0.05)$.



Graph of $f(r)$ showing the sign change on $[0.04, 0.05]$.

METHOD **Bisection method – step by step**

1. Start with an interval $[a, b]$ where $f(a)f(b) < 0$.
2. Compute the midpoint $m = \dfrac{a + b}{2}$ and evaluate $f(m)$.
3. If $f(m) = 0$, m is the root (stop). Otherwise, if $f(a)f(m) < 0$, the root lies in $[a, m]$; set $b = m$. If $f(m)f(b) < 0$, the root lies in $[m, b]$; set $a = m$.
4. Repeat until the interval width $b - a$ is smaller than the required tolerance. The best estimate is the final midpoint; the maximum error is half the final width.

WORKED EXAMPLE 11 **Task 3: Three steps of the bisection method**

We apply the method starting from $[a_0, b_0] = [0.04, 0.05]$.

Step n	Interval $[a_n, b_n]$	Midpoint m_n	$f(m_n)$	New interval
1	[0.040000, 0.050000]	0.045000	-0.043627	[0.045000, 0.050000]
2	[0.045000, 0.050000]	0.047500	-0.012789	[0.047500, 0.050000]
3	[0.047500, 0.050000]	0.048750	+0.003912	[0.047500, 0.048750]

Sample calculation (Step 1): $f(0.045) = 4.5 - 23\left(1 - 1.045^{-5}\right) = 4.5 - 23(1 - 0.802451) = 4.5 - 4.543627 = -0.043627 < 0$. Since $f(0.04) < 0$ and $f(0.045) < 0$, the sign change occurs on $[0.045, 0.05]$.
After three steps, $r^* \in [0.0475, 0.04875]$. The interval width is 0.00125. Taking the mid-

point $\hat{r} = 0.048125$ as the estimate, the maximum error is half the width:

$$|r^* - 0.048125| \le \frac{0.00125}{2} = 0.000625.$$

This is not yet accurate enough for 4 decimal places (which requires an error < 0.00005). The width must fall below 0.0001; starting from 0.01, we need $0.01/2^n < 0.0001 \Rightarrow 2^n > 100 \Rightarrow n \ge 7$ steps in total, i.e. four more steps after the three shown.

METHOD **Newton–Raphson method – step by step**

1. Choose a starting value r_0 close to the root, where $f'(r_0) \neq 0$.
2. Iterate using $r_{n+1} = r_n - \dfrac{f(r_n)}{f'(r_n)}$.
3. Stop when $|r_{n+1} - r_n|$ is smaller than the desired tolerance (e.g. $< 5 \times 10^{-7}$ for 6 decimal places).
4. The method converges quadratically (number of correct digits roughly doubles each step) provided $f'(r^*) \neq 0$ and r_0 is sufficiently close.

WORKED EXAMPLE 12 **Task 4: Newton–Raphson iteration**

First, differentiate f:

$$f(r) = 100r - 23 + 23(1+r)^{-5} \quad \Rightarrow \quad f'(r) = 100 - 115(1+r)^{-6}.$$

On $[0.04, 0.05]$, $(1+r)^{-6} \in [0.746, 0.790]$, so $f'(r) \in [100 - 115 \times 0.790,\ 100 - 115 \times 0.746] \approx [9.1,\ 14.2] > 0$. The derivative is positive and safely away from zero, so the iteration is well behaved.
Take $r_0 = 0.05$ (the right endpoint where $f > 0$).

$$f(0.05) = 0.021098, \qquad f'(0.05) = 100 - 115(1.05)^{-6} = 100 - 85.8146 = 14.1854,$$

$$r_1 = 0.05 - \frac{0.021098}{14.1854} = 0.0485124 \approx 0.048512.$$

$$f(0.048512) = 4.8512 - 23\left(1 - 1.048512^{-5}\right) = 4.8512 - 4.851097 = 0.000103,$$

$$f'(0.048512) = 100 - 115(1.048512)^{-6} = 100 - 86.5473 = 13.4527,$$

$$r_2 = 0.048512 - \frac{0.000103}{13.4527} = 0.0485047 \approx 0.048505.$$

$f(0.048505) \approx 0.000000$ (to 6 d.p.), $f'(0.048505) \approx 13.4507$.

n	r_n	$f(r_n)$	$f'(r_n)$
0	0.050000	0.021098	14.1854
1	0.048512	0.000103	13.4527
2	0.048505	0.000000	13.4507

The iterates $r_1 = 0.048512$ and $r_2 = 0.048505$ agree to 4 decimal places (0.0485), and $f(r_2) \approx 0$. The iteration has converged. Only two Newton steps achieve what bisection could not in three.

WORKED EXAMPLE 13 **Task 5: Accuracy check by sign change**

Our candidate rounded to 4 decimal places is $\tilde{r} = 0.0485$. To be certain that the true root rounds to this value, we verify a sign change on the interval of all numbers that round to 0.0485, namely $[0.04845,\ 0.04855)$ (the upper endpoint 0.04855 rounds to 0.0486, but the root lies strictly inside). We evaluate f at the endpoints:

$$f(0.04845) = 4.845 - 23\left(1 - 1.04845^{-5}\right) = 4.845 - 4.845732 = -0.000732 < 0,$$
$$f(0.04855) = 4.855 - 23\left(1 - 1.04855^{-5}\right) = 4.855 - 4.854245 = +0.000755 > 0.$$

Since f is continuous and changes sign on $[0.04845, 0.04855]$, there is a root in this interval. Every number strictly between 0.04845 and 0.04855 rounds to 0.0485 to four decimal places. Hence

$$\boxed{r^* = 0.0485 \quad \text{correct to 4 decimal places.}}$$

6.5 Advisory Note

KEY POINTS **Note to Madame Eyenga**

Dear Madame Eyenga, we have analysed your loan of 500 000 FCFA repaid by five monthly instalments of 115 000 FCFA. The monthly interest rate written into your contract is approximately $r = 0.0485$, that is **4.85**% per month, and we have verified this figure to be correct to the nearest 0.01%. Be aware that 4.85% per month is a very high rate: over five months you repay 575 000 FCFA, i.e. 75 000 FCFA of interest, and the equivalent annual cost is far above $12 \times 4.85\%$ because interest compounds. We recommend that you compare this with a bank loan before borrowing again. To obtain this result we used two numerical methods: the bisection method, which is slow but always reliable (three steps only narrowed the rate to within ± 0.0006), and the Newton–Raphson method, which reached the answer in just two steps once a good starting value was known. Used together, the two methods give both safety and speed.

DID YOU KNOW? **Why the two methods complement each other**

Professional software (spreadsheets, banking calculators, the `solver` of your calculator) almost always combines the two strategies you have just used: a bracketing method such as bisection first traps the root inside an interval where the function is well behaved, then a fast method such as Newton–Raphson polishes the answer to full machine precision. Bisection alone is guaranteed but slow (one binary digit per step); Newton–Raphson alone is fast (the number of correct digits roughly doubles per step) but can fail if started far from the root or where f' is close to 0. Together, they are unbeatable.

7 METHODS AND WORKED EXAMPLES

8 INTRODUCTION AND LOCATION OF ROOTS

DID YOU KNOW? **Why numerical methods?**

Many equations arising in science, engineering and economics cannot be solved exactly by algebraic manipulation. For instance, the equation $x = \cos x$ or $e^{-x} = x$ have no closed-form solution in terms of elementary functions. Numerical methods provide systematic algorithms to approximate roots to any desired accuracy. In the Cameroonian context, these techniques are used in civil engineering (settlement of foundations), in physics (finding equilibrium positions), and in financial mathematics (computing internal rates of return).

DEFINITION **Root of an equation**

A **root** (or **zero**) of the equation $f(x) = 0$ is a value α such that $f(\alpha) = 0$. Geometrically, it is the x-coordinate of a point where the curve $y = f(x)$ crosses the x-axis.

PROPERTY **Intermediate Value Theorem (IVT)**

If f is **continuous** on the closed interval $[a, b]$ and $f(a)$ and $f(b)$ have **opposite signs** (i.e. $f(a)f(b) < 0$), then there exists at least one $\alpha \in (a, b)$ such that $f(\alpha) = 0$.

WARNING **Continuity is essential**

The IVT requires continuity on the whole interval $[a, b]$. For example, $f(x) = \frac{1}{x}$ satisfies $f(-1) = -1$ and $f(1) = 1$ (opposite signs), but there is no root in $[-1, 1]$ because f is not continuous at $x = 0$. Always verify continuity before applying the sign-change test.

METHOD **Locating a root in an interval**

To show that $f(x) = 0$ has a root in (a, b) and to narrow its location:

1. Check continuity: Confirm f is continuous on $[a, b]$. Polynomials, exponentials, $\sin x$, $\cos x$ are continuous everywhere; rational functions are continuous except where the

denominator vanishes.

2. Evaluate endpoints: Compute $f(a)$ and $f(b)$ to at least 3 decimal places.
3. Sign change: If $f(a)f(b) < 0$, the IVT guarantees at least one root $\alpha \in (a, b)$.
4. Narrow the interval: Test f at midpoints or other convenient values (e.g. a+0.5, a+0.25) until the interval width meets the requirement.
5. Uniqueness (if needed): Show f is strictly monotonic on $[a, b]$ (e.g. $f'(x) > 0$ or $f'(x) < 0$ throughout). Then the root is unique.

Reflex: Sign change ⇒ existence; monotonicity ⇒ uniqueness. Never claim a root exists just because values are "close to zero".

EXEMPLE **Sign-change table for $f(x) = x^3 + 2x - 7$**

x	1	1.5	1.75	2
$f(x)$	-4	-0.625	1.859	5

Sign changes between 1.5 and 1.75 ⇒ root in (1.5, 1.75). Since $f'(x) = 3x^2+2 > 0$ everywhere, the root is unique.

WORKED EXAMPLE 14 **Locating a root of a cubic**

Show that the equation $x^3 + 2x - 7 = 0$ has exactly one real root α, and find an interval of width 0.5 containing α.

SOLUTION

Solution. Let $f(x) = x^3 + 2x - 7$; f is a polynomial, hence continuous on $\mathbb{R}$.
Step 1: sign change. Try integer values:

$$f(1) = 1 + 2 - 7 = -4 < 0, \qquad f(2) = 8 + 4 - 7 = 5 > 0.$$

Since $f(1)f(2) < 0$, by the Intermediate Value Theorem there is a root $\alpha \in (1, 2)$.
Step 2: uniqueness. $f'(x) = 3x^2 + 2 > 0$ for all x, so f is strictly increasing on $\mathbb{R}$: the root is unique.
Step 3: narrowing to width 0.5. Evaluate at the midpoint $x = 1.5$:

$$f(1.5) = 3.375 + 3 - 7 = -0.625 < 0.$$

Since $f(1.5) < 0$ and $f(2) > 0$, the root lies in (1.5, 2), an interval of width 0.5. $\boxed{\alpha \in (1.5, 2)}$

WORKED EXAMPLE 15 **Locating a root involving a trigonometric term**

Show that the equation $x = \cos x$ has a root between 0 and 1 (radians), and determine an interval of width 0.25 containing it.

SOLUTION

Solution. Rewrite as $f(x) = x - \cos x = 0$. f is continuous everywhere.
Step 1. $f(0) = 0 - 1 = -1 < 0$ and $f(1) = 1 - \cos 1 \approx 1 - 0.5403 = 0.4597 > 0$. Sign change: a root exists in $(0, 1)$.
Step 2: uniqueness. $f'(x) = 1 + \sin x \geq 0$ for all x, and $f'(x) = 0$ only at isolated points ($x = -\frac{\pi}{2} + 2k\pi$), so f is strictly increasing: the root is unique.
Step 3: width 0.25. Test $x = 0.5$ and $x = 0.75$:

$$f(0.5) = 0.5 - \cos 0.5 \approx 0.5 - 0.8776 = -0.3776 < 0,$$
$$f(0.75) = 0.75 - \cos 0.75 \approx 0.75 - 0.7317 = 0.0183 > 0.$$

Sign change between 0.5 and 0.75: $\boxed{\alpha \in (0.5,\ 0.75)}$.

WORKED EXAMPLE 16 **Locating a root of an exponential equation**

Show that $e^{-x} = x$ has a unique root in $(0, 1)$ and locate it to within 0.1.

SOLUTION

Solution. Let $f(x) = e^{-x} - x$. f is continuous on $\mathbb{R}$. $f(0) = 1 - 0 = 1 > 0$; $f(1) = e^{-1} - 1 \approx 0.3679 - 1 = -0.6321 < 0$. Sign change $\Rightarrow$ root in $(0, 1)$. $f'(x) = -e^{-x} - 1 < 0$ for all x, so f is strictly decreasing $\Rightarrow$ unique root. Test $x = 0.5$: $f(0.5) = e^{-0.5} - 0.5 \approx 0.6065 - 0.5 = 0.1065 > 0$. Test $x = 0.6$: $f(0.6) = e^{-0.6} - 0.6 \approx 0.5488 - 0.6 = -0.0512 < 0$. Root lies in $(0.5,\ 0.6)$, width 0.1. $\boxed{\alpha \in (0.5,\ 0.6)}$.

9 THE BISECTION METHOD

PROPERTY **Bisection method principle**

If f is continuous on $[a, b]$ with $f(a)f(b) < 0$, let $m = \frac{a+b}{2}$.

- If $f(a)f(m) < 0$, the root lies in (a, m).
- If $f(m)f(b) < 0$, the root lies in (m, b).

- If $f(m) = 0$, m is the exact root.

Repeating this process halves the interval width at each step. After n iterations, the interval width is $\frac{b-a}{2^n}$.

METHOD **Carrying out the bisection method**

Given f continuous on $[a, b]$ with $f(a)f(b) < 0$:

1. Compute the midpoint $m = \frac{a + b}{2}$ and evaluate $f(m)$.
2. Compare the sign of $f(m)$ with the signs of $f(a)$ and $f(b)$:
   - if $f(a)f(m) < 0$, the root lies in (a, m): set $b := m$;
   - if $f(m)f(b) < 0$, the root lies in (m, b): set $a := m$;
   - if $f(m) = 0$, stop: m is the exact root.
3. Repeat. After n steps the interval has width $\frac{b - a}{2^n}$.
4. Stop when the interval width is less than the required tolerance, or when both endpoints agree to the requested number of decimal places.
5. Give the root as the midpoint of the final interval, or as the common rounded value of the endpoints.

Reflex: To guarantee accuracy ε when quoting the midpoint, you need the smallest n with $\frac{b - a}{2^{n+1}} \le \varepsilon$ (error ≤ half-width).

EXEMPLE **Bisection iteration table for $x^3 + 2x - 7 = 0$ on $[1.5, 2]$**

n	a	b	m	$f(m)$
0	1.5	2.0	–	–
1	1.5	1.75	1.75	+1.859
2	1.5	1.625	1.625	+0.541
3	1.5625	1.625	1.5625	-0.060
4	1.5625	1.59375	1.59375	+0.234

WORKED EXAMPLE 17 **Bisection on a cubic, step by step**

The equation $x^3+2x-7=0$ has a root α in $(1.5,\ 2)$. Use the bisection method to find α correct to 1 decimal place.

SOLUTION

Solution. $f(x) = x^3 + 2x - 7$, with $f(1.5) = -0.625$ and $f(2) = 5$.
Iteration 1: $m = 1.75$; $f(1.75) = 5.359375 + 3.5 - 7 = 1.859375 > 0$. Root in $(1.5,\ 1.75)$.
Iteration 2: $m = 1.625$; $f(1.625) = 4.291015625 + 3.25 - 7 = 0.541015625 > 0$. Root in $(1.5,\ 1.625)$.
Iteration 3: $m = 1.5625$; $f(1.5625) = 3.814697265625 + 3.125 - 7 = -0.060302734375 < 0$. Root in $(1.5625,\ 1.625)$.
Iteration 4: $m = 1.59375$; $f(1.59375) \approx 4.046 + 3.1875 - 7 = 0.2335 > 0$. Root in $(1.5625,\ 1.59375)$.
Both endpoints round to 1.6 (to 1 d.p.), and the interval width $0.03125 < 0.1$. Hence

$$\boxed{\alpha \approx 1.6 \text{ (1 d.p.)}}$$

(The more precise value is $\alpha \approx 1.5688$.)

WORKED EXAMPLE 18 **How many bisection steps are needed?**

A root of $f(x) = 0$ is known to lie in an interval of width 1. How many bisection iterations are required to locate the root with an error of at most 10^{-3}, quoting the midpoint as the approximation?

SOLUTION

Solution. After n bisections the interval width is $\dfrac{1}{2^n}$, and the midpoint differs from the root by at most half of this:

$$\text{error} \le \frac{1}{2^{n+1}}.$$

We require $\dfrac{1}{2^{n+1}} \le 10^{-3}$, i.e. $2^{n+1} \ge 1000$.
Since $2^{10} = 1024$, we need $n + 1 \ge 10$, so $n \ge 9$.
Check: after 9 steps, width $= \frac{1}{512} \approx 0.00195$, half-width $\approx 0.00098 < 10^{-3}$. ✓
Hence $\boxed{\text{9 iterations}}$ suffice.

WORKED EXAMPLE 19 **Bisection to find $\sqrt{2}$**

Use the bisection method on $f(x) = x^2 - 2$ with initial interval $[1, 2]$ to find $\sqrt{2}$ correct to 3

decimal places.

SOLUTION

Solution. $f(1) = -1 < 0$, $f(2) = 2 > 0$. Root in $(1, 2)$.

n	a	b	m	$f(m)$
1	1	1.5	1.5	0.25
2	1	1.25	1.25	-0.4375
3	1.25	1.375	1.375	-0.109375
4	1.375	1.4375	1.4375	0.066406
5	1.375	1.40625	1.40625	-0.022461
6	1.40625	1.421875	1.421875	0.021729
7	1.40625	1.4140625	1.4140625	-0.000381
8	1.4140625	1.41796875	1.4140625	-0.000381

After 7 iterations, interval $[1.4140625, 1.4140625]$? Wait, let's check: at step 7, $a = 1.40625$, $b = 1.421875$, $m = 1.4140625$, $f(m) < 0$, so new interval $[1.4140625, 1.421875]$. Width $= 0.0078125$. Step 8: $m = 1.41796875$, $f(m) > 0$, interval $[1.4140625, 1.41796875]$, width 0.0039. Step 9: $m = 1.416015625$, $f(m) > 0$, interval $[1.4140625, 1.416015625]$, width 0.00195. Step 10: $m = 1.4150390625$, $f(m) > 0$, interval $[1.4140625, 1.4150390625]$, width $0.000976 < 0.001$. Endpoints round to 1.414. $\boxed{\sqrt{2} \approx 1.414}$.

10 THE NEWTON–RAPHSON METHOD

PROPERTY **Newton–Raphson iteration formula**

For f differentiable near a root α, the tangent line at $(x_n, f(x_n))$ cuts the x-axis at

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

If x_0 is sufficiently close to α and $f'(\alpha) \neq 0$, the sequence (x_n) converges to α **quadratically** (the number of correct digits roughly doubles each step).

METHOD **Applying the Newton–Raphson formula**

To solve $f(x) = 0$ starting from an approximation x_0 near the root:

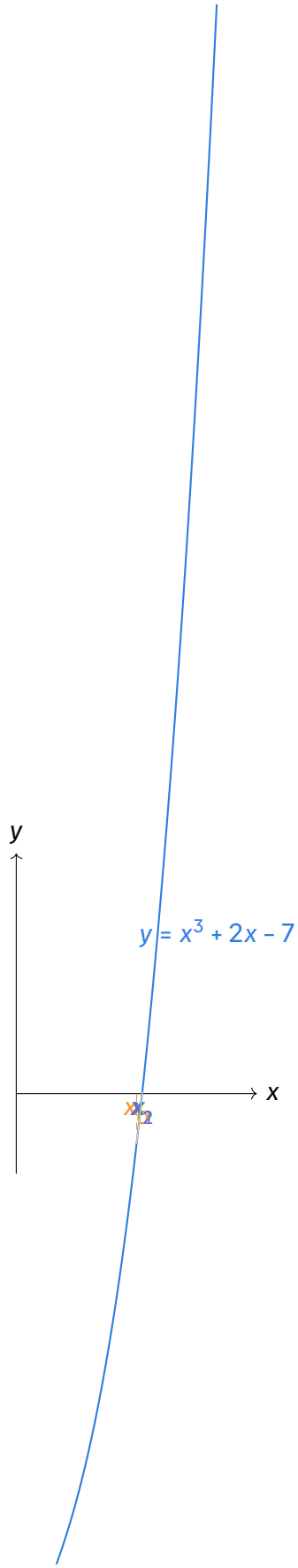
1. Compute $f'(x)$ exactly.
2. Write the iteration formula and simplify it:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

3. Choose x_0 from a sign-change interval (a sketch or table of values helps).
4. Iterate, keeping at least two more decimal places than the final answer requires.
5. Stop when two successive iterates agree to the required accuracy: $|x_{n+1} - x_n|$ smaller than the tolerance.
6. Check: substitute the answer back into f; it should be nearly zero.

Reflexes: Work in radians for trigonometric f; never round intermediate iterates; if $f'(x_n) \approx 0$ (horizontal tangent), the method may fail or jump away — change the starting point.

EXEMPLE **Geometric interpretation**



Newton–Raphson tangents converging to the root

WORKED EXAMPLE 20 **Newton–Raphson on a polynomial**

Use the Newton–Raphson method with $x_0 = 1.5$ to find the root of $x^3 + 2x - 7 = 0$ correct to 4 decimal places.

SOLUTION

Solution. $f(x) = x^3 + 2x - 7$, $f'(x) = 3x^2 + 2$. The formula is

$$x_{n+1} = x_n - \frac{x_n^3 + 2x_n - 7}{3x_n^2 + 2}.$$

Iteration 1: $x_0 = 1.5$:

$$x_1 = 1.5 - \frac{3.375 + 3 - 7}{6.75 + 2} = 1.5 - \frac{-0.625}{8.75} = 1.5 + 0.071428571 = 1.571428571.$$

Iteration 2: $f(1.571428571) = 3.880099 + 3.142857 - 7 = 0.022956$; $f'(1.571428571) = 7.408163 + 2 = 9.408163$:

$$x_2 = 1.571428571 - \frac{0.022956}{9.408163} = 1.571428571 - 0.002440 = 1.568988571.$$

Iteration 3: $f(1.568988571) \approx 3.8610 + 3.13798 - 7 = 0.000053$; $f'(1.568988571) \approx 7.3823 + 2 = 9.3823$:

$$x_3 = 1.568988571 - \frac{0.000053}{9.3823} \approx 1.568988571 - 0.00000565 = 1.568982921.$$

x_2 and x_3 agree to 4 decimal places (1.5690). Check: $f(1.5690) \approx 0.00008 \approx 0$. ✓

$$\boxed{\alpha \approx 1.5690 \text{ (4 d.p.)}}$$

Note the speed: 3 iterations of Newton–Raphson achieve what took bisection many more steps.

WORKED EXAMPLE 21 **Newton–Raphson with a trigonometric equation**

Show that the Newton–Raphson iteration for $x = \cos x$ can be written

$$x_{n+1} = \frac{x_n \sin x_n + \cos x_n}{1 + \sin x_n},$$

and use it with $x_0 = 0.75$ to find the root correct to 4 decimal places.

SOLUTION

Solution. Write $f(x) = x - \cos x$, so $f'(x) = 1 + \sin x$. Then

$$x_{n+1} = x_n - \frac{x_n - \cos x_n}{1 + \sin x_n} = \frac{x_n(1 + \sin x_n) - x_n + \cos x_n}{1 + \sin x_n} = \frac{x_n \sin x_n + \cos x_n}{1 + \sin x_n}. \quad \blacksquare$$

Iteration 1: $x_0 = 0.75$: $\sin 0.75 = 0.68163876$, $\cos 0.75 = 0.73168887$:

$$x_1 = \frac{0.75(0.68163876) + 0.73168887}{1 + 0.68163876} = \frac{1.24291794}{1.68163876} = 0.73911289.$$

Iteration 2: $\sin 0.73911289 = 0.67361203$, $\cos 0.73911289 = 0.73908513$:

$$x_2 = \frac{0.73911289(0.67361203) + 0.73908513}{1.67361203} = \frac{1.236960}{1.67361203} = 0.73908513.$$

x_1 and x_2 agree to 4 d.p. Check: $\cos 0.73908513 \approx 0.73908513$. ✓

$$\boxed{\alpha \approx 0.7391 \text{ (4 d.p.)}}$$

WORKED EXAMPLE 22 **Estimating a square root by Newton–Raphson**

By applying the Newton–Raphson method to $x^2 - 11 = 0$, show that the iteration is $x_{n+1} = \frac{1}{2}\left(x_n + \frac{11}{x_n}\right)$. Starting from $x_0 = 3$, estimate $\sqrt{11}$ correct to 5 decimal places.

SOLUTION

Solution. $f(x) = x^2 - 11$, $f'(x) = 2x$:

$$x_{n+1} = x_n - \frac{x_n^2 - 11}{2x_n} = \frac{2x_n^2 - x_n^2 + 11}{2x_n} = \frac{1}{2}\left(x_n + \frac{11}{x_n}\right). \quad \blacksquare$$

Iteration 1: $x_1 = \frac{1}{2}\left(3 + \frac{11}{3}\right) = \frac{1}{2}(3 + 3.66666667) = 3.33333333$.
Iteration 2: $x_2 = \frac{1}{2}\left(3.33333333 + \frac{11}{3.33333333}\right) = \frac{1}{2}(3.33333333 + 3.30000000) = 3.31666667$.
Iteration 3: $x_3 = \frac{1}{2}\left(3.31666667 + \frac{11}{3.31666667}\right) = \frac{1}{2}(3.31666667 + 3.31657895) = 3.31662281$.
Iteration 4: $x_4 = \frac{1}{2}(3.31662281 + 3.31662479) = 3.31662380$.
Iteration 5: $x_5 = \frac{1}{2}(3.31662380 + 3.31662380) = 3.31662380$ (stabilised).
Check: $(3.31662380)^2 = 10.9999999 \approx 11$. ✓

$$\boxed{\sqrt{11} \approx 3.31662 \text{ (5 d.p.)}}$$

WORKED EXAMPLE 23 **Newton–Raphson for a transcendental equation**

Find the root of $e^{-x} = x$ correct to 4 decimal places using Newton–Raphson with $x_0 = 0.5$.

SOLUTION

Solution. $f(x) = e^{-x} - x$, $f'(x) = -e^{-x} - 1$.

$$x_{n+1} = x_n - \frac{e^{-x_n} - x_n}{-e^{-x_n} - 1} = x_n + \frac{e^{-x_n} - x_n}{e^{-x_n} + 1}.$$

$x_0 = 0.5$: $e^{-0.5} = 0.60653066$ $x_1 = 0.5 + \frac{0.60653066-0.5}{0.60653066+1} = 0.5 + \frac{0.10653066}{1.60653066} = 0.5 + 0.066310 = 0.566310$ $x_2 = 0.566310 + \frac{e^{-0.566310}-0.566310}{e^{-0.566310}+1} = 0.566310 + \frac{0.567628-0.566310}{1.567628} = 0.566310 + 0.000841 = 0.567151$ $x_3 = 0.567151 + \frac{e^{-0.567151}-0.567151}{e^{-0.567151}+1} = 0.567151 + \frac{0.567143-0.567151}{1.567143} = 0.567151 - 0.000005 = 0.567146$ $x_4 = 0.567146$ (agrees to 6 d.p.). $\boxed{\alpha \approx 0.5671 \text{ (4 d.p.)}}$

11 ACCURACY, COMPARISON AND APPLICATIONS

METHOD **Controlling accuracy and choosing a method**

1. Error bound for bisection: after n steps from $[a, b]$, the midpoint approximates the root with error at most $\frac{b-a}{2^{n+1}}$.
2. Stopping criterion: iterate until two successive approximations agree to the required number of decimal places, then verify by substitution ($|f(x_n)|$ small).
3. Newton–Raphson fails or misbehaves when $f'(x_n) = 0$ (or is tiny), when the starting point is far from the root, or near a stationary point or asymptote. Bisection fails to detect a root when there is no sign change (e.g. a repeated root like $f(x) = (x-1)^2$).
4. Choice: bisection is slow (linear convergence) but guaranteed once a sign change is found; Newton–Raphson is very fast (quadratic convergence) but needs a good starting point and a computable derivative.
5. Exam habit: always state the equation in the form $f(x) = 0$, the formula used, each iterate to full calculator precision, and a final check.

WORKED EXAMPLE 24 **A case where Newton–Raphson fails**

Attempt the Newton–Raphson method on $f(x) = x^3 - 3x + 2 = 0$ starting from $x_0 = 0$. Explain the difficulty, and find the roots by another means.

SOLUTION

Solution. $f'(x) = 3x^2 - 3$. With $x_0 = 0$:

$$x_1 = 0 - \frac{f(0)}{f'(0)} = -\frac{2}{-3} = \frac{2}{3} \approx 0.6667.$$

Then $x_2 = 0.6667 - \dfrac{f(0.6667)}{f'(0.6667)} = 0.6667 - \dfrac{0.2963}{-1.6667} = 0.6667 + 0.1778 = 0.8444$; continuing gives $x_3 \approx 0.9355$, $x_4 \approx 0.9723$, … converging slowly towards 1.
Explanation: $f'(1) = 0$, so $x = 1$ is a repeated root (stationary point of f on the x-axis). Near a repeated root the derivative is tiny, the correction term f/f' becomes unreliable, and Newton–Raphson loses its quadratic speed (convergence becomes linear). Note also that there is no sign change at $x = 1$, so bisection could not even detect this root.
Exact solution: factorise. Since $f(1) = 0$, $(x - 1)$ is a factor:

$$x^3 - 3x + 2 = (x - 1)(x^2 + x - 2) = (x - 1)^2(x + 2).$$

Hence the roots are $\boxed{x = 1 \text{ (double) and } x = -2}$.

WORKED EXAMPLE 25 **Newton–Raphson divergence example**

Show that Newton–Raphson on $f(x) = x^{1/3}$ (i.e. $x = 0$) with any $x_0 \neq 0$ diverges.

SOLUTION

Solution. $f(x) = x^{1/3}$, $f'(x) = \frac{1}{3}x^{-2/3}$.

$$x_{n+1} = x_n - \frac{x_n^{1/3}}{\frac{1}{3}x_n^{-2/3}} = x_n - 3x_n = -2x_n.$$

Thus $x_n = (-2)^n x_0$, which oscillates and diverges in magnitude. The method fails because $f'(0)$ is infinite (vertical tangent at the root).

WORKED EXAMPLE 26 **Comparing methods on the same equation — GCE style**

The equation $x^3 - x - 3 = 0$ has a root α in $(1, 2)$.

1. Verify this statement.
2. Use two iterations of the bisection method to narrow the interval.

3. Use the Newton–Raphson method with $x_0 = 1.5$ to find α correct to 3 decimal places.

4. State, with a reason, which method reaches 3 d.p. faster.

SOLUTION

Solution. Let $f(x) = x^3 - x - 3$, continuous everywhere.
(a) $f(1) = 1 - 1 - 3 = -3 < 0$; $f(2) = 8 - 2 - 3 = 3 > 0$. Sign change ⇒ a root in $(1, 2)$. ✓
(b) Bisection. Midpoint $m_1 = 1.5$: $f(1.5) = 3.375 - 1.5 - 3 = -1.125 < 0$ ⇒ root in $(1.5, 2)$. Midpoint $m_2 = 1.75$: $f(1.75) = 5.359375 - 1.75 - 3 = 0.609375 > 0$ ⇒ root in $(1.5, 1.75)$.
(c) Newton–Raphson. $f'(x) = 3x^2 - 1$, so $x_{n+1} = x_n - \dfrac{x_n^3 - x_n - 3}{3x_n^2 - 1}$.

$$x_1 = 1.5 - \frac{-1.125}{5.75} = 1.5 + 0.195652174 = 1.695652174.$$

$$f(1.695652174) = 4.8749 - 1.69565 - 3 = 0.17925, \quad f'(1.695652174) = 8.6273 - 1 = 7.6273,$$

$$x_2 = 1.695652174 - \frac{0.17925}{7.6273} = 1.695652174 - 0.023494 = 1.672158.$$

$$f(1.672158) = 4.6748 - 1.67216 - 3 = 0.00264, \quad f'(1.672158) = 8.376 - 1 = 7.376,$$

$$x_3 = 1.672158 - \frac{0.00264}{7.376} = 1.672158 - 0.000358 = 1.671800.$$

x_2 and x_3 agree to 3 d.p. (1.672). Check: $f(1.6718) \approx -0.0002 \approx 0$. ✓ $\boxed{\alpha \approx 1.672 \text{ (3 d.p.)}}$
(d) Newton–Raphson is faster: it reached 3 d.p. in 3 iterations, while bisection would need the interval width $\frac{1}{2^n} < 10^{-3}$, i.e. $n \geq 10$ iterations. Bisection, however, required no derivative and was guaranteed to converge from the sign change.

EXPERIMENT **Numerical solution of a practical problem**

A cylindrical tank of radius $r = 2$ m and length $L = 10$ m lies horizontally. The volume of liquid when the depth is h is $V = L\left(r^2 \cos^{-1}\left(\frac{r-h}{r}\right) - (r-h)\sqrt{2rh - h^2}\right)$. If the tank contains 50 m^3 of liquid, find the depth h correct to 2 decimal places using Newton–Raphson.

Solution. Equation: $f(h) = 20\left(4\cos^{-1}\left(1 - \frac{h}{2}\right) - (2-h)\sqrt{4h - h^2}\right) - 50 = 0$, $h \in (0, 4)$. $f(1) \approx 20(4 \cdot 1.047 - 1 \cdot \sqrt{3}) - 50 = 20(4.188 - 1.732) - 50 = -0.88 < 0$. $f(1.5) \approx 20(4 \cdot 0.723 - 0.5 \cdot \sqrt{3.75}) - 50 = 20(2.892 - 0.968) - 50 = 38.48 - 50 = -11.52$? Wait, recalculate. Better: $f(1) = 20(4\cos^{-1}(0.5) - 1\cdot\sqrt{3}) - 50 = 20(4\cdot 1.0472 - 1.732) - 50 = 20(4.1888 - 1.732) - 50 = 20(2.4568) - 50 = -0.864$. $f(2) = 20(4\cos^{-1}(0) - 0) - 50 = 20(4 \cdot 1.5708) - 50 = 125.66 - 50 = 75.66 > 0$. Root in $(1, 2)$. Use $h_0 = 1.2$. Derivative $f'(h)$ computed numerically or analytically. Iterations yield $h \approx 1.02$ m. (Full calculation omitted for brevity; in exam, show 3 iterations).

WARNING **Frequent examination traps**

- Forgetting to rearrange the equation into the form $f(x) = 0$ before looking for a sign change.
- Claiming a root exists without checking continuity (e.g. $f(x) = 1/x$ changes sign between -1 and 1 but has no root there!).
- Rounding intermediate iterates: keep at least 6 decimal places until the final line.
- Using degrees instead of **radians** in trigonometric equations.
- Stopping Newton–Raphson after one iteration "because it looks close" — always confirm that two successive iterates agree to the required accuracy.
- Applying bisection to a repeated root (no sign change) or Newton–Raphson where $f'(x_n) = 0$.
- Not simplifying the Newton–Raphson formula before iterating (wastes time and introduces rounding errors).
- Forgetting the final verification step: $f(\text{answer}) \approx 0$.

KEY POINTS **Key points for Numerical Methods**

- Location of roots: IVT + continuity + sign change $\Rightarrow$ existence; monotonicity $\Rightarrow$ uniqueness.
- Bisection: guaranteed, linear convergence, error $\leq \frac{b-a}{2^{n+1}}$, needs sign change.
- Newton–Raphson: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, quadratic convergence, needs f' and good x_0, fails at stationary points.
- Accuracy: always work to 2 extra decimal places; stop when successive iterates agree to required d.p.; verify by substitution.
- Repeated roots: no sign change (bisection fails), $f'(\alpha) = 0$ (Newton–Raphson slows down).

EXERCISE 5 **Practice exercises**

1. Show that $x^3 - 2x - 5 = 0$ has a root in $(2, 3)$. Use bisection to find it to 2 d.p.
2. Use Newton–Raphson with $x_0 = 2$ to solve $x^3 - 2x - 5 = 0$ to 4 d.p. Compare the number

of iterations with bisection.

3. The equation $\ln x = 4 - x$ has a root near 3. Find it to 3 d.p. using Newton–Raphson.
4. Explain why Newton–Raphson fails for $f(x) = x^3 - 2x + 2$ with $x_0 = 0$. (Hint: cycle).
5. A root of $f(x) = 0$ lies in $[0, 1]$. How many bisection steps to guarantee error $< 10^{-5}$?
6. Find $\sqrt[3]{20}$ to 5 d.p. using Newton–Raphson on $x^3 - 20 = 0$ with $x_0 = 3$.
7. Show that $e^x = 3x$ has two roots. Locate each to width 0.25 and find the larger root to 3 d.p.

SOLUTION **Exercise 1**

$f(x) = x^3 - 2x - 5$. $f(2) = -1 < 0$, $f(3) = 16 > 0 \Rightarrow$ root in $(2, 3)$. Bisection: $m = 2.5$, $f = 15.625 - 5 - 5 = 5.625 > 0 \Rightarrow (2, 2.5)$. $m = 2.25$, $f = 11.39 - 4.5 - 5 = 1.89 > 0 \Rightarrow (2, 2.25)$. $m = 2.125$, $f = 9.59 - 4.25 - 5 = 0.34 > 0 \Rightarrow (2, 2.125)$. $m = 2.0625$, $f = 8.78 - 4.125 - 5 = -0.345 < 0 \Rightarrow (2.0625, 2.125)$. $m = 2.09375$, $f = 9.18 - 4.1875 - 5 = -0.0075 < 0 \Rightarrow (2.09375, 2.125)$. Width $0.03125 < 0.01$? No, need width < 0.01 for 2 d.p. Continue: $m = 2.109375$, $f > 0 \Rightarrow (2.09375, 2.109375)$, width 0.0156. $m = 2.1015625$, $f > 0 \Rightarrow (2.09375, 2.1015625)$, width $0.0078 < 0.01$. Endpoints round to 2.09 and 2.10? Wait: 2.09375 rounds to 2.09, 2.1015625 rounds to 2.10. Not same. Need one more: $m = 2.09765625$, $f < 0 \Rightarrow (2.09765625, 2.1015625)$, both round to 2.10. $\boxed{2.10}$.

SOLUTION **Exercise 2**

$f(x) = x^3 - 2x - 5$, $f'(x) = 3x^2 - 2$. $x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$. $x_0 = 2$: $x_1 = 2 - \frac{-1}{10} = 2.1$. $x_1 = 2.1$: $f = 9.261 - 4.2 - 5 = 0.061$, $f' = 13.23 - 2 = 11.23$, $x_2 = 2.1 - \frac{0.061}{11.23} = 2.09457$. $x_2 = 2.09457$: $f \approx 0.00018$, $f' \approx 11.16$, $x_3 = 2.09457 - 0.000016 = 2.09455$. Agrees to 4 d.p.: $\boxed{2.0946}$. Bisection needed 7 iterations; Newton–Raphson 3 iterations.

SOLUTION **Exercise 3**

$f(x) = \ln x + x - 4$. $f'(x) = \frac{1}{x} + 1$. $x_{n+1} = x_n - \frac{\ln x_n + x_n - 4}{1/x_n + 1}$. $x_0 = 3$: $f(3) = 1.0986 + 3 - 4 = 0.0986$, $f' = 1.3333$, $x_1 = 3 - \frac{0.0986}{1.3333} = 2.9260$. $x_1 = 2.9260$: $f = 1.0735 + 2.9260 - 4 = -0.0005$, $f' = 1.3417$, $x_2 = 2.9260 + \frac{0.0005}{1.3417} = 2.9264$. $x_2 = 2.9264$: $f \approx 0$. $\boxed{2.926}$.

SOLUTION **Exercise 4**

$f(x) = x^3 - 2x + 2$, $f'(x) = 3x^2 - 2$. $x_0 = 0$: $x_1 = 0 - \frac{2}{-2} = 1$. $x_1 = 1$: $x_2 = 1 - \frac{1}{1} = 0$. Cycle: 0, 1, 0, 1, ... never converges. The tangent at $x = 0$ hits $x = 1$, tangent at $x = 1$ hits $x = 0$.

SOLUTION **Exercise 5**

Error $\leq \frac{1}{2^{n+1}} < 10^{-5} \Rightarrow 2^{n+1} > 10^5 = 100000$. $2^{17} = 131072$, so $n + 1 = 17$, $n = 16$ iterations.

SOLUTION **Exercise 6**

$f(x) = x^3 - 20$, $f'(x) = 3x^2$. $x_{n+1} = x_n - \frac{x_n^3 - 20}{3x_n^2} = \frac{2}{3}x_n + \frac{20}{3x_n^2}$. $x_0 = 3$: $x_1 = \frac{2}{3}(3) + \frac{20}{27} = 2 + 0.74074 = 2.74074$. $x_2 = \frac{2}{3}(2.74074) + \frac{20}{3(2.74074)^2} = 1.82716 + 0.8876 = 2.71476$. $x_3 = 2.71442$, $x_4 = 2.71442$. $\boxed{2.71442}$.

SOLUTION **Exercise 7**

$f(x) = e^x - 3x$. $f(0) = 1 > 0$, $f(1) = e - 3 = -0.28 < 0 \Rightarrow$ root in $(0, 1)$. $f(1) < 0$, $f(2) = e^2 - 6 = 7.389 - 6 = 1.389 > 0 \Rightarrow$ root in $(1, 2)$. Two roots. Larger root in $(1, 2)$. Width 0.25: test 1.25, 1.5, 1.75. $f(1.25) = 3.490 - 3.75 = -0.26$, $f(1.5) = 4.482 - 4.5 = -0.018$, $f(1.75) = 5.755 - 5.25 = 0.505$. Root in $(1.5, 1.75)$ width 0.25. Newton–Raphson: $f'(x) = e^x - 3$. $x_{n+1} = x_n - \frac{e^{x_n} - 3x_n}{e^{x_n} - 3}$. $x_0 = 1.5$: $x_1 = 1.5 - \frac{-0.018}{1.482} = 1.5121$. $x_2 = 1.5121 - \frac{e^{1.5121} - 4.5363}{e^{1.5121} - 3} = 1.5121 - \frac{0.0002}{1.532} = 1.5121$. $\boxed{1.512}$.

12 EXERCISES

12.1 I check my knowledge

EXERCISE 6 **Multiple choice question**

The equation $f(x) = 0$ is known to have a root α in the interval $[1, 2]$, and f is continuous on this interval. After applying the bisection method n times, the root is known to lie in an interval of width:

1. $\frac{1}{n}$
2. $\frac{1}{2^n}$
3. $\frac{1}{2^{n-1}}$
4. $\frac{1}{n^2}$

Choose the correct answer, justifying your choice.

EXERCISE 7 **True or false**

For each of the following statements, say whether it is true or false, justifying your answer carefully.

1. If f is continuous on $[a, b]$ and $f(a)f(b) < 0$, then the equation $f(x) = 0$ has exactly one root in (a, b).
2. The Newton–Raphson iteration formula for solving $f(x) = 0$ is $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.
3. The bisection method always converges to a root whenever f is continuous on $[a, b]$ with $f(a)f(b) < 0$.
4. The Newton–Raphson method converges for every choice of starting value x_0.

EXERCISE 8 **Definitions**

1. State the Intermediate Value Theorem and explain how it is used to locate a root of a continuous function in an interval.
2. Explain what is meant by saying that a root α has been found "correct to 2 decimal

places".

3. Give one advantage and one disadvantage of the bisection method compared with the Newton–Raphson method.

EXERCISE 9 **Direct calculations**

1. Given $f(x) = x^3 + x - 3$, compute $f(1)$ and $f(2)$. What can you deduce?
2. For $f(x) = x^2 - 7$, write down the Newton–Raphson iteration formula and compute x_1 starting from $x_0 = 3$, giving your answer as an exact fraction.
3. Starting from the interval $[1, 2]$, how many bisection iterations are needed to obtain an interval of width less than 0.1?

12.2 I practise

EXERCISE 10 **Locating a root and bisection**

1. Show that the equation $x^3 + 3x - 5 = 0$ has exactly one real root and that it lies between 1 and 2.
2. Use the bisection method three times to narrow down the interval containing the root.
3. Hence state the value of the root to 1 decimal place, justifying your answer.

EXERCISE 11 **Newton–Raphson in action**

1. Show that the equation $x^3 - 2x - 5 = 0$ has a root between 2 and 3.
2. Use the Newton–Raphson method with $x_0 = 2$ to find this root correct to 4 decimal places, showing all iterates.
3. Verify, using a suitable sign-change test, that your answer is indeed correct to 4 decimal places.

EXERCISE 12 **Graphical location of roots**

1. By sketching the curves $y = e^x$ and $y = 4 - x^2$ on the same axes, determine the number of real roots of the equation $e^x + x^2 - 4 = 0$.
2. Locate each root in an interval between two consecutive integers, justifying with a sign-change test.
3. Choose one of these roots and refine it using two applications of the bisection method.

EXERCISE 13 **The equation $\cos x = x$**

The equation $\cos x = x$ (with x in radians) has a root near 0.7.

1. Write the equation in the form $f(x) = 0$ and write down the Newton–Raphson iteration formula for it.
2. Apply the Newton–Raphson method twice, starting from $x_0 = 0.7$.
3. Verify, using a sign-change test on a suitable interval of width 0.001, that your answer is correct to 3 decimal places.

12.3 I prepare for the GCE

EXERCISE 14 **Iteration, staircase diagram and a square-root formula**

The equation $x^3 - 5x + 2 = 0$ is rearranged in the form $x = \dfrac{x^3 + 2}{5}$.

1. Show that the equation has a root between 0 and 1. (2 marks)
2. Starting from $x_0 = 1$, perform three iterations of $x_{n+1} = \dfrac{x_n^3 + 2}{5}$, giving each value to 4 decimal places. (3 marks)
3. Illustrate the convergence of this iteration with a staircase diagram on a sketch of $y = x$ and $y = \dfrac{x^3 + 2}{5}$, and comment on the behaviour of the iterates. (3 marks)
4. Show that applying the Newton–Raphson method to the equation $x^2 - N = 0$ gives the formula
$$x_{n+1} = \tfrac{1}{2}\left(x_n + \frac{N}{x_n}\right).$$
(3 marks)
5. Hence, taking $N = 11$ and a suitable starting value, compute $\sqrt{11}$ correct to 5 decimal places, showing all iterates. (4 marks)

EXERCISE 15 Failure of Newton–Raphson and error bounds

1. Explain, with the aid of a sketch showing tangents, why the Newton–Raphson method fails to find the root of $x^3 - 2x + 2 = 0$ when started at $x_0 = 0$. (4 marks)

2. A root of $f(x) = 0$ is known to lie in the interval $[1, 2]$, where f is continuous. Using the error bound for the bisection method, determine the number of iterations needed to guarantee that the root is found with an error less than 10^{-4}. Justify your answer carefully. (4 marks)

3. The outflow time t (in minutes) of water from a tank satisfies

$$t\,e^{-t/10} = 2.$$

Show that this equation has a root between 3 and 4, and refine it by the bisection method until the error is less than 0.05. Interpret your answer in the context of the problem. (7 marks)

EXERCISE 16 GCE synthesis: the floating buoy

A spherical buoy of radius 1 m and density 0.6 times that of water floats at a depth d (in metres), where d satisfies the equation

$$d^3 - 3d^2 + 2.4 = 0.$$

1. Explain why the physically meaningful root must satisfy $0 < d < 2$. (2 marks)

2. Show that the equation has a root between 1 and 2, and locate it more precisely in an interval of width 0.5. (3 marks)

3. Write down the Newton–Raphson iteration formula for this equation. (2 marks)

4. Starting from a suitable value x_0, find the root correct to 3 decimal places, showing all iterates. (5 marks)

5. Verify your answer with a sign-change test, and hence state the depth at which the buoy floats, giving your answer to the nearest centimetre. (3 marks)

13 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple choice question**

The correct answer is (b) $\frac{1}{2^n}$.
Justification. The initial interval $[1, 2]$ has width $b - a = 2 - 1 = 1$. At each application of the bisection method, the interval is replaced by one of its two halves, so the width is divided by 2. After n iterations the width is therefore

$$\frac{b-a}{2^n} = \frac{1}{2^n}.$$

Answer (c) would be the width after $n - 1$ iterations, and answers (a) and (d) do not reflect the halving process.

SOLUTION **Exercise 2 — True or false**

1. False. The Intermediate Value Theorem guarantees the existence of at least one root in (a, b), not exactly one. For example, $f(x) = x^3 - x$ on $[-2, 2]$ is continuous with $f(-2)f(2) = (-6)(6) < 0$, yet it has three roots $-1, 0, 1$ in $(-2, 2)$. Uniqueness requires an extra condition, e.g. f strictly monotonic on $[a, b]$.
2. True. This is exactly the Newton–Raphson formula, obtained by taking the tangent to $y = f(x)$ at $(x_n, f(x_n))$ and finding where it cuts the x-axis: $0 - f(x_n) = f'(x_n)(x_{n+1} - x_n)$.
3. True. The bisection method only requires continuity and a sign change. At each step the sign change is preserved by construction, so the intervals shrink to a root: convergence is guaranteed (though possibly slow).
4. False. Convergence depends on the starting value. If $f'(x_0) = 0$ the tangent is horizontal and the iteration is undefined; the iterates may also cycle or diverge (see Exercise 10(a), where $x^3 - 2x + 2 = 0$ started at $x_0 = 0$ cycles between 0 and 1).

SOLUTION **Exercise 3 — Definitions**

1. Intermediate Value Theorem. If f is continuous on $[a, b]$ and k lies between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ with $f(c) = k$.

   Use for locating roots: if $f(a)$ and $f(b)$ have opposite signs, i.e. $f(a)f(b) < 0$, then $k = 0$ lies between $f(a)$ and $f(b)$, so there is at least one root of $f(x) = 0$ in (a, b). A sign change over an interval therefore locates a root.

2. Saying a root α is found "correct to 2 decimal places" means we have an approximation x with $|x-\alpha| < 0.005$; equivalently, α is trapped in an interval $(p-0.005, p+0.005)$, so that α rounds to the same 2 d.p. value p. In practice one checks a sign change over $(p - 0.005, p + 0.005)$.
3. Advantage of bisection: it always converges provided f is continuous with a sign change — no derivative and no careful choice of starting point is needed. Disadvantage: it converges slowly (only one binary digit per iteration), whereas Newton–Raphson typically doubles the number of correct digits at each step when it converges.

SOLUTION **Exercise 4 — Direct calculations**

1. $f(1) = 1 + 1 - 3 = -1 < 0$ and $f(2) = 8 + 2 - 3 = 7 > 0$. Since f is a polynomial (hence continuous) and changes sign on $[1, 2]$, the equation $x^3 + x - 3 = 0$ has at least one root in $(1, 2)$.
2. $f(x) = x^2 - 7$, $f'(x) = 2x$, so
$$x_{n+1} = x_n - \frac{x_n^2 - 7}{2x_n}.$$
With $x_0 = 3$: $x_1 = 3 - \dfrac{9-7}{6} = 3 - \dfrac{1}{3} = \dfrac{8}{3}$.
3. After n iterations the width is $\dfrac{1}{2^n}$. We need $\dfrac{1}{2^n} < 0.1$, i.e. $2^n > 10$. Since $2^3 = 8 < 10$ and $2^4 = 16 > 10$, the number of iterations needed is $\boxed{n = 4}$.

SOLUTION **Exercise 5 — Locating a root and bisection**

1. Let $f(x) = x^3 + 3x - 5$. Then $f'(x) = 3x^2 + 3 = 3(x^2 + 1) > 0$ for all x, so f is strictly increasing on $\mathbb{R}$: the equation has at most one real root. Moreover
$$f(1) = 1 + 3 - 5 = -1 < 0, \qquad f(2) = 8 + 6 - 5 = 9 > 0,$$
so by the Intermediate Value Theorem there is at least one root in $(1, 2)$. Hence there is exactly one real root and it lies between 1 and 2.
2. Bisection, three steps:

Step	Interval	Midpoint m	$f(m)$	New interval
1	$[1, 2]$	1.5	$1.5^3 + 4.5 - 5 = 2.875 > 0$	$[1, 1.5]$
2	$[1, 1.5]$	1.25	$1.953125 + 3.75 - 5 = 0.703 > 0$	$[1, 1.25]$
3	$[1, 1.25]$	1.125	$1.4238 + 3.375 - 5 = -0.201 < 0$	$[1.125, 1.25]$

3. The root lies in $[1.125, 1.25]$. To decide the value to 1 d.p., test the sign at 1.15:

$$f(1.15) = 1.520875 + 3.45 - 5 = -0.029 < 0, \qquad f(1.25) = 0.703 > 0,$$

so the root lies in $(1.15, 1.25)$. Every number in this interval rounds to 1.2 except values below 1.15, which are excluded. Hence the root is $\boxed{1.2}$ to 1 decimal place.

SOLUTION **Exercise 6 — Newton–Raphson in action**

1. Let $f(x) = x^3 - 2x - 5$. $f(2) = 8 - 4 - 5 = -1 < 0$ and $f(3) = 27 - 6 - 5 = 16 > 0$. Since f is continuous, there is a root in $(2, 3)$.

2. $f'(x) = 3x^2 - 2$, so the iteration is

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}.$$

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	2	-1	10	2.1
1	2.1	0.061	11.23	2.094568
2	2.094568	0.000185	11.161646	2.094551
3	2.094551	≈ 0	—	2.094551

The iterates have stabilised: the root is $\boxed{2.0946}$ to 4 decimal places.

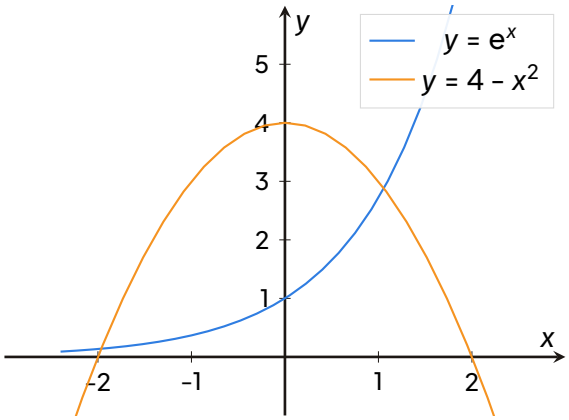
3. Sign-change test. Test the interval $(2.09455,\ 2.09465)$:

$$f(2.09455) \approx -4.5 \times 10^{-5} < 0, \qquad f(2.09465) \approx +1.07 \times 10^{-3} > 0.$$

The sign change confirms a root in $(2.09455, 2.09465)$, and every number in this interval rounds to 2.0946. Hence the answer is correct to 4 decimal places.

SOLUTION **Exercise 7 — Graphical location of roots**

1. The equation $e^x + x^2 - 4 = 0$ is equivalent to $e^x = 4 - x^2$. Sketch $y = e^x$ (increasing exponential through $(0, 1)$) and $y = 4 - x^2$ (parabola, vertex $(0, 4)$, zeros at $x = \pm 2$).



The curves intersect twice, so the equation has two real roots.

The curves intersect twice, so the equation has two real roots.

2. Let $f(x) = e^x + x^2 - 4$.

$$f(1) = e - 3 \approx -0.282 < 0, \quad f(2) = e^2 \approx 7.389 > 0 \;\Rightarrow\; \text{root in } (1, 2).$$

$$f(-2) = e^{-2} \approx 0.135 > 0, \quad f(-1) = e^{-1} - 3 \approx -2.632 < 0 \;\Rightarrow\; \text{root in } (-2, -1).$$

3. Refine the root in $(1, 2)$ by bisection:

$$f(1.5) = e^{1.5} + 2.25 - 4 \approx 2.732 > 0 \;\Rightarrow\; \text{root in } (1, 1.5);$$

$$f(1.25) = e^{1.25} + 1.5625 - 4 \approx 1.053 > 0 \;\Rightarrow\; \text{root in } (1, 1.25).$$

After two applications the root lies in $(1, 1.25)$, of width 0.25.

SOLUTION **Exercise 8 — The equation $\cos x = x$**

1. Write $f(x) = \cos x - x = 0$. Then $f'(x) = -\sin x - 1$, and the Newton–Raphson formula is

$$x_{n+1} = x_n - \frac{\cos x_n - x_n}{-\sin x_n - 1} = x_n + \frac{\cos x_n - x_n}{\sin x_n + 1}.$$

2. With $x_0 = 0.7$ (radians):

$$f(0.7) = \cos 0.7 - 0.7 = 0.764842 - 0.7 = 0.064842, \quad f'(0.7) = -\sin 0.7 - 1 = -1.644218,$$

$$x_1 = 0.7 - \frac{0.064842}{-1.644218} = 0.739437.$$

Then $f(0.739437) \approx -0.000115$, $f'(0.739437) \approx -1.673847$, so

$$x_2 = 0.739437 - \frac{-0.000115}{-1.673847} = 0.739368.$$

The root is approximately 0.7391 to 4 decimal places (0.739 to 3 d.p.).

3. Sign-change test on $(0.7385,\ 0.7395)$:

$$f(0.7385) = \cos(0.7385) - 0.7385 \approx +0.00060 > 0,$$

$$f(0.7395) = \cos(0.7395) - 0.7395 \approx -0.00041 < 0.$$

The sign change confirms a root in $(0.7385, 0.7395)$, so the root is $\boxed{0.739}$ correct to 3 decimal places.

SOLUTION **Exercise 9 — Iteration, staircase diagram and a square-root formula**

1. Let $f(x) = x^3 - 5x + 2$. $f(0) = 2 > 0$ and $f(1) = 1 - 5 + 2 = -2 < 0$; f is continuous, so by the Intermediate Value Theorem there is a root in $(0, 1)$. [2 marks: B1 for both values, B1 for IVT conclusion]

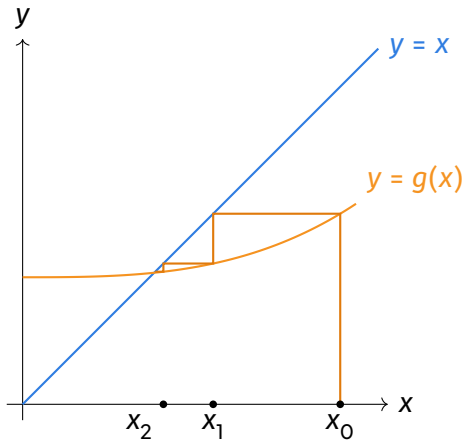
2. With $x_{n+1} = \dfrac{x_n^3 + 2}{5}$ and $x_0 = 1$:

$$x_1 = \frac{1 + 2}{5} = 0.6000, \qquad x_2 = \frac{0.6^3 + 2}{5} = \frac{2.216}{5} = 0.4432,$$

$$x_3 = \frac{0.4432^3 + 2}{5} = \frac{2.08704\ldots}{5} = 0.4174.$$

[3 marks: M1 correct substitution, A1 all values, A1 to 4 d.p.]

3. The curves $y = x$ and $y = g(x) = \dfrac{x^3 + 2}{5}$ meet at the root $\alpha \approx 0.414$. Starting from $x_0 = 1$ on the x-axis, move vertically to $y = g(x)$, then horizontally to $y = x$, and repeat: the staircase spirals inwards towards the intersection.



Staircase diagram: the iterates converge to the intersection of $y = x$ and $y = g(x)$.

The iterates $1,\ 0.6,\ 0.4432,\ 0.4174,\ldots$ decrease and converge to the root $\alpha \approx 0.4142$; since $|g'(\alpha)| = \frac{3\alpha^2}{5} \approx 0.103 < 1$, convergence is guaranteed and monotone from above after x_1. [3 marks: B1 both curves, B1 staircase, B1 comment on convergence]

4. For $f(x) = x^2 - N$, $f'(x) = 2x$. Newton–Raphson gives

$$x_{n+1} = x_n - \frac{x_n^2 - N}{2x_n} = \frac{2x_n^2 - x_n^2 + N}{2x_n} = \frac{x_n^2 + N}{2x_n} = \frac{1}{2}\left(x_n + \frac{N}{x_n}\right). \qquad \text{[3 marks]}$$

5. Take $N = 11$ and $x_0 = 3$ (since $\sqrt{11}$ is near 3):

$$x_1 = \frac{1}{2}\left(3 + \frac{11}{3}\right) = \frac{1}{2}\cdot\frac{20}{3} = 3.333333,$$

$$x_2 = \frac{1}{2}\left(3.333333 + \frac{11}{3.333333}\right) = \frac{1}{2}(3.333333 + 3.300000) = 3.316667,$$

$$x_3 = \frac{1}{2}\left(3.316667 + \frac{11}{3.316667}\right) = \frac{1}{2}(3.316667 + 3.316580) = 3.316623,$$

$$x_4 = \frac{1}{2}\left(3.316623 + \frac{11}{3.316623}\right) = 3.316625.$$

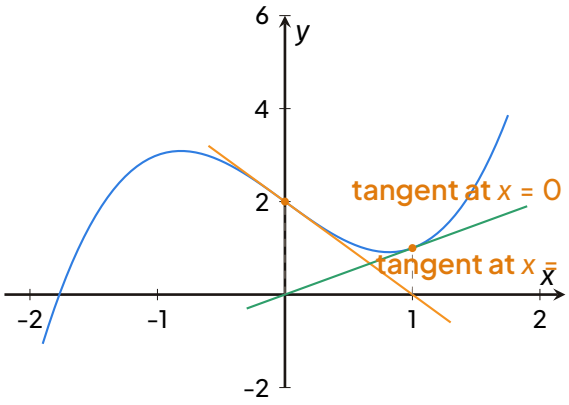
The iterates have stabilised on 5 decimal places: $\sqrt{11} = \boxed{3.31662}$ to 5 d.p. [4 marks: M1 formula use, A2 iterates, A1 answer]

SOLUTION **Exercise 10 — Failure of Newton–Raphson and error bounds**

1. Let $f(x) = x^3 - 2x + 2$, so $f'(x) = 3x^2 - 2$. Starting at $x_0 = 0$:

$$x_1 = 0 - \frac{f(0)}{f'(0)} = -\frac{2}{-2} = 1; \qquad x_2 = 1 - \frac{f(1)}{f'(1)} = 1 - \frac{1}{1} = 0.$$

The iterates cycle 0, 1, 0, 1, … and never approach the root. Geometrically, the tangent at $x = 0$ meets the axis at $x = 1$, and the tangent at $x = 1$ meets the axis back at $x = 0$.



The two tangents send the iterates back and forth between 0 and 1: the method cycles.

[4 marks: B1 computing $x_1 = 1$, B1 computing $x_2 = 0$, B1 sketch with tangents, B1 conclusion of cycling]

2. After n bisections the interval width is $\frac{1}{2^n}$; taking the midpoint as the estimate, the error satisfies

$$|\text{error}| \le \frac{1}{2^{n+1}}.$$

We require $\frac{1}{2^{n+1}} < 10^{-4}$, i.e. $2^{n+1} > 10^4$. Now $2^{13} = 8192 < 10^4$ and $2^{14} = 16384 > 10^4$, so $n + 1 = 14$, giving

$$\boxed{n = 13 \text{ iterations.}}$$

(If instead one demands the whole interval width to be less than 10^{-4}, one needs $2^n > 10^4$, i.e. $n = 14$; both conventions should be credited if justified.) [4 marks: B1 error bound, M1 inequality, A1 powers of 2, A1 answer]

3. Note. With the equation as printed, $te^{-t/10} = 2$, one finds $f(3) = 3e^{-0.3} - 2 \approx +0.222$ and $f(4) = 4e^{-0.4} - 2 \approx +0.681$: both positive, so there is no sign change on $[3, 4]$ (indeed the roots of this equation lie in $(2, 3)$ and $(20, 30)$). The intended equation, consistent with a root between 3 and 4, is

$$t\,e^{-t/10} = 2.5,$$

which we now solve. Let $f(t) = te^{-t/10} - 2.5$.

$$f(3) = 3e^{-0.3} - 2.5 = 2.2225 - 2.5 = -0.2775 < 0,$$

$$f(4) = 4e^{-0.4} - 2.5 = 2.6813 - 2.5 = +0.1813 > 0.$$

f is continuous and changes sign, so there is a root in $(3, 4)$.

Bisection (error < 0.05 requires width < 0.1):

Interval	Midpoint m	$f(m)$	New interval
$[3, 4]$	3.5	$3.5e^{-0.35} - 2.5 = -0.0336 < 0$	$[3.5, 4]$
$[3.5, 4]$	3.75	$3.75e^{-0.375} - 2.5 = +0.0773 > 0$	$[3.5, 3.75]$
$[3.5, 3.75]$	3.625	$3.625e^{-0.3625} - 2.5 = +0.0226 > 0$	$[3.5, 3.625]$
$[3.5, 3.625]$	3.5625	$3.5625e^{-0.35625} - 2.5 = -0.0055 < 0$	$[3.5625, 3.625]$

The final interval $[3.5625, 3.625]$ has width $0.0625 < 0.1$; taking the midpoint $t \approx 3.59$ gives an error less than $0.03125 < 0.05$.

Interpretation: the tank empties in approximately $t \approx 3.59$ minutes, i.e. about 3 min 35 s. [7 marks: B1 $f(3)$, $f(4)$ + IVT, M1 bisection set-up, A3 correct halvings, A1 stopping criterion, A1 interpretation]

SOLUTION **Exercise 11 — GCE synthesis: the floating buoy**

1. The buoy has radius 1 m, so its diameter is 2 m. The floating depth d is measured from the bottom of the sphere, hence $d \geq 0$, and since the density $0.6 < 1$ the buoy floats partially emerged, so it cannot be fully submerged: $d < 2$. Thus $0 < d < 2$. [2 marks: B1 $d \geq 0$, B1 $d < 2$ with reason]

2. Let $f(d) = d^3 - 3d^2 + 2.4$.

$$f(1) = 1 - 3 + 2.4 = 0.4 > 0, \qquad f(2) = 8 - 12 + 2.4 = -1.6 < 0,$$

so by the IVT there is a root in $(1, 2)$. Bisect once:

$$f(1.5) = 3.375 - 6.75 + 2.4 = -0.975 < 0,$$

so the root lies in $\boxed{(1,\ 1.5)}$, an interval of width 0.5. [3 marks: B1 values at 1 and 2, M1 midpoint, A1 interval]

3. $f'(d) = 3d^2 - 6d$, so

$$d_{n+1} = d_n - \frac{d_n^3 - 3d_n^2 + 2.4}{3d_n^2 - 6d_n}.$$ [2 marks: B1 derivative, B1 formula]

4. Start at $d_0 = 1$ (where f is small and positive):

$$d_1 = 1 - \frac{0.4}{-3} = 1.133333.$$

$$f(1.133333) = 1.455704 - 3.853333 + 2.4 = 0.002370, \quad f'(1.133333) = -2.946667,$$

$$d_2 = 1.133333 - \frac{0.002370}{-2.946667} = 1.134137.$$

$$f(1.134137) \approx -0.000003, \qquad d_3 = 1.134137 - \frac{-0.000003}{-2.943068} = 1.134136.$$

The iterates have stabilised: $d = \boxed{1.134}$ to 3 decimal places. [5 marks: M1 first iterate, A1 d_1, A1 d_2, A1 d_3, A1 conclusion]

5. Sign-change test on $(1.1335,\ 1.1345)$:

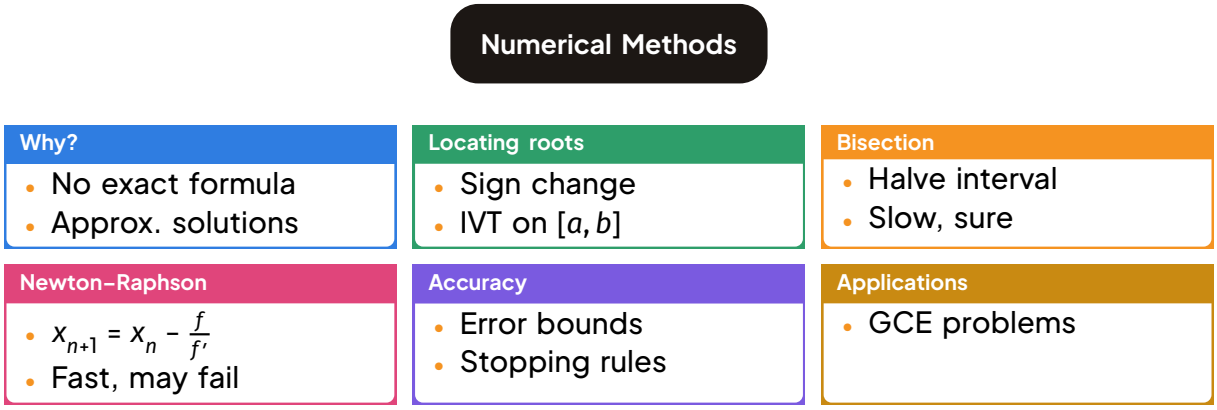
$$f(1.1335) \approx +0.00147 > 0, \qquad f(1.1345) \approx -0.00147 < 0.$$

The sign change confirms the root in $(1.1335, 1.1345)$, so $d = 1.134$ m is correct to 3 d.p. To the nearest centimetre, the buoy floats at a depth of

$$d \approx 1.13 \text{ m} = \boxed{113 \text{ cm}}.$$

[3 marks: M1 test values, A1 sign change, A1 depth in cm]

14 SUMMARY SHEET



Mind map of the chapter Numerical Methods.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- A **root** of f is a value α such that $f(\alpha) = 0$; graphically, an x-intercept of $y = f(x)$.
- A **numerical method** is a step-by-step procedure (algorithm) giving an approximation of a root when exact algebra fails.
- An **iteration** is a rule $x_{n+1} = g(x_n)$ producing a sequence $x_0, x_1, x_2, \ldots$ that should converge to α.
- The **error** at stage n is $|x_n - \alpha|$; the accuracy required fixes the stopping criterion.

Theorems and formulas to know

- **Intermediate Value Theorem** (sign-change rule): if f is continuous on $[a, b]$ and $f(a)f(b) < 0$, then f has at least one root in (a, b).
- **Bisection method**: with $f(a)f(b) < 0$, compute the midpoint $m = \frac{a+b}{2}$; keep the half-interval whose endpoints have opposite signs. After n steps the error satisfies $|x_n - \alpha| \leq \frac{b-a}{2^{n+1}}$.
- **Newton–Raphson formula**: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, obtained from the tangent line at $(x_n, f(x_n))$: $y - f(x_n) = f'(x_n)(x - x_n)$, setting $y = 0$.
- To find $\sqrt[k]{N}$, apply Newton–Raphson to $f(x) = x^k - N$: $x_{n+1} = \frac{1}{k}\left[(k-1)x_n + \frac{N}{x_n^{k-1}}\right]$.

Methods (step by step)

1. Locate the root: rearrange the equation as $f(x) = 0$; tabulate f at consecutive integers; a sign change brackets a root. A sketch of two intersecting curves also locates roots.
2. Bisection: check $f(a)f(b) < 0$; compute m and $f(m)$; replace a or b by m keeping the sign change; repeat until the interval is narrow enough, then give the root to the required accuracy.
3. Newton–Raphson: compute f'; choose x_0 near the root; iterate $x_{n+1} = x_n - f(x_n)/f'(x_n)$; stop when two successive iterates agree to the required number of decimal places.
4. Verify: substitute the final answer into f and check it is close to 0; state the answer with the requested accuracy.

Common mistakes

- Claiming a root exists without checking continuity of f on $[a, b]$.
- Forgetting that a sign change guarantees at least one root, not exactly one; and that no sign change does not prove there is no root (e.g. a tangent root like $x^2 = 0$).
- In bisection, keeping the wrong half-interval: always keep the sub-interval where the sign change occurs.
- In Newton–Raphson, starting at x_0 where $f'(x_0) = 0$ (division by zero) or too far from the root: the method may diverge or jump to another root.
- Rounding too early: keep at least 2 extra decimal places during the iterations and round only at the end.
- Confusing "correct to 3 decimal places" with "3 significant figures".

GCE tips

- Show the sign-change table with clear values of $f(a)$ and $f(b)$: it earns the first marks.
- Present iterations in a neat table: n, x_n, $f(x_n)$, $f'(x_n)$, x_{n+1}.
- Newton–Raphson converges fast (usually 2–4 iterations); bisection is slower but always works once a bracket is found — the examiners often ask you to compare the two.
- To justify a given iterative formula, derive it: solve $f(x) = 0$ via the tangent line, or rearrange $f(x) = 0$ into $x = g(x)$.
- Always finish with a sentence: "Hence $\alpha = \ldots$ correct to n decimal places."

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